

Supplementary Materials: Prediction-Driven Non-Preemptive Scheduling with Bounded Overtaking for Shared Execution Services under Deadline-Driven Bursty Load

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This file holds supporting material for the manuscript: the visibility audit of the feature pipeline, the predictor families and the predictor scores on the two judging traces and on the four non-judging ones (Section S1); the field audit of the main judging trace, what supports the shared-pool design, two checks on the continuous-integration refit and the cost-concentration screen (Section S2); the experimental configuration, the selected budget points, the budget shapes the main guard table no longer prints, the corrupted scores, the aging comparator, the alternative ranking losses, the identity residual on all nine cells, the single-server configuration and the trace-construction sensitivities (Section S3); and the deferred proofs and remarks of the theory section, the workload comparison and the consistency proposition among them, including the proof of the net-overtake identity (Section S4). Section numbers here carry an S prefix and every table is numbered S. A cross-reference without an S — “Section 8.2”, “Table 5”, “Theorem 1” — points into the main manuscript, since a \ref across the two files does not resolve. The traces are the ones the manuscript’s Table 2 names: the CodeBench judging logs [1], the Azure Functions invocation trace [2], the Intel Netbatch logs [3] and the two Firefox continuous-integration pools collected from Mozilla’s public APIs [4,5]. The additional reproducibility specification is Section S5; capacity, physical queueing, feedback and LPC-EGEE results are Section S6. Additional theory is Section S7; the extremal constructions and remaining proofs behind Section 6 of the manuscript, which carries no appendix of its own, are Section S8; and the per-paper detail behind Section 2 of the manuscript is Section S9. Every empirical figure quoted here is a development figure in the sense of Section 8.2 of the manuscript: nothing in this file comes from a sealed part of any trace, and nothing here was recomputed for it.

S1. Cost Prediction

S1.1. The Visibility Audit in Full

Section 4.2 of the manuscript states that every historical feature is recomputed by streaming the log in availability order and reading each entity’s state as of the query instant, and that 0 rows disagree in any of the eight configurations. This is what that test is.

On 3,000 randomly drawn development jobs all 34 history and relational features are recomputed from scratch under each configuration’s own visibility rule and compared against the pipeline’s values. The same test applied to the natural but wrong construction, in which a feature may use any record with a smaller index, fails on 97.67% of rows under the primary rule, so the test does discriminate. The graph pipeline is checked separately: an index assertion over 863,149 rows finds 0 timeline entries pointing at or after the read, brute-force recomputation on 600 rows matches on all 2,400 root states and all 2,400 neighbour sets, and perturbing or deleting every record after a cut time leaves all earlier predictions unchanged while a positive control moves the later ones.

S1.2. Predictors on the Non-Judging Traces

Table S1 is the cross-trace counterpart of Table S2. The three figures Section 4.3 of the manuscript quotes from it are the serverless STATIC Spearman correlation and the two CI

pools' ENT AUROC; the AUROC column of Table 7 of the manuscript carries the figure the scheduling results there are read against, which is the ALL row of each trace.

Table S1. The same protocol on the non-judging traces, by feature group. ENT uses the entity's own visible history, STATIC uses only what is declared before the job runs, ALL is both. RMSE and Spearman are on $\log(1 + C)$.

Trace	Features	AUROC	AUPRC	RMSE	Spearman
Serverless invocations	ENT	0.9828	0.816	0.3401	0.953
	STATIC	0.6238	0.084	0.8088	0.021
	ALL	0.9847	0.830	0.3185	0.932
CI pool, hardware A	ENT	0.6334	0.157	0.3964	0.833
	STATIC	0.4990	0.045	0.7659	0.277
	ALL	0.6512	0.177	0.3956	0.837
CI pool, hardware B	ENT	0.9968	0.984	0.0887	0.985
	STATIC	0.6915	0.077	0.6411	0.500
	ALL	0.9970	0.985	0.0860	0.988
Compute farm	ENT	0.8903	0.265	1.4214	0.883
	STATIC	0.8704	0.242	2.2559	0.750
	ALL	0.8902	0.260	1.3485	0.892

S1.3. The Release Delay as a Sensitivity

This subsection supports the third commitment of Section 4.2 of the manuscript, that the delay with which results reach the feature store is treated as a sensitivity and not as an assumption.

Third, we treat the delay δ with which results reach the feature store as a sensitivity and not as an assumption. At $\delta = 0$ the full tabular model reaches AUROC 0.9283 and at $\delta = 600$ s it reaches 0.9066. On the second judging platform, where the log carries no clock and lag has to be expressed in records, the same model moves from 0.9266 at lag 0 to 0.8902 at lag 1,000. Ranking survives a ten-minute feedback delay at the top of the curve; retrieval of the small heavy class degrades, average precision falling from 0.289 to 0.139.

S1.4. Predictor Families

This subsection defines the three model families the manuscript compares in Section 4.3 under one visibility protocol and one split, and the ablation ladder its negative relational result is read off.

All families see the same rows, the same split and the same availability rule. The tabular family is gradient-boosted trees [6] over five feature groups: task history (M1), user history (M2), static features of the submitted artefact (M3), everything tabular (M4), and M4 plus one-hop relational aggregates over the task-assignment and user-cohort neighbourhoods (M5). A control (M6) permutes task identity across assignments, so that M5's aggregates are keyed on a scrambled neighbourhood; it reaches AUROC 0.9288 in the primary configuration against M5's 0.9302, so the aggregates lose almost nothing when the identity they are built on is destroyed. The sequence family is a GRU over the earlier jobs of one user whose results are already available, used alone (R1) and as a state vector appended to M4's columns inside the boosted model (R1S). The relational family is a heterogeneous network over user, task, assignment and cohort nodes [7] with no node-identity embeddings, so an unseen task or user is handled by the same weights. Its ablation ladder runs N0 (a multilayer perceptron on exactly M4's columns), G0U (the relational architecture with every neighbour slot empty and the assignment and cohort roots removed, which leaves exactly M4's information), G0 (empty neighbour slots, roots kept), G1 (the real edges) and G2 (the same network on edges permuted within a period and time block, so that neighbour counts and recency are preserved and only the identity of the neighbour changes).

S1.5. Predictors on the Two Judging Traces

This subsection carries the table behind Section 4.3 of the manuscript, the predictor scores on the two judging traces.

Table S2. Predictors on the two judging traces under one visibility protocol and one split. Heavy is the training-period 95th percentile of executed cost. RMSE and Spearman (Sp.) are on $\log(1 + C)$. R-net is the heterogeneous relational network.

Model	Information	AUROC	AUPRC	RMSE	Sp.
<i>CodeBench</i> , $n = 40,844$					
M1	task history	0.7960	0.053	0.3000	0.219
M2	user history	0.8591	0.242	0.2605	0.302
M3	static features	0.8632	0.082	0.2892	0.290
M4	all tabular	0.9283	0.289	0.2519	0.312
M5	M4 + one-hop	0.9302	0.299	0.2507	0.344
N0	MLP on M4's columns	0.9209	0.306	0.2463	0.278
G0U	R-net, no edges or roots	0.9348	0.322	0.2417	0.321
G0	R-net, no edges	0.9325	0.316	0.2402	0.324
G1	R-net, real edges	0.9324	0.322	0.2397	0.329
G2	R-net, permuted edges	0.9305	0.318	0.2396	0.323
R1	GRU over user history	0.9239	0.324	0.2396	0.313
R1S	M4 columns + GRU state	0.9366	0.315	0.2414	0.329
<i>ACcoding</i> , $n = 647,465$; the neural models were not fitted here					
M1	task history	0.8990	0.431	0.2904	0.683
M2	user history	0.7855	0.292	0.3190	0.471
M3	static features	0.7884	0.165	0.3429	0.493
M4	all tabular	0.9266	0.527	0.2714	0.692
M5	M4 + one-hop	0.9264	0.527	0.2713	0.696

S1.6. The Cost of Predicting

This subsection supports the manuscript's statement in Section 4.4 that the choice between these predictors is a queueing decision and not a latency one.

Prediction has to be cheap enough to sit in front of a queue whose median job is 204 ms. Measured in a streaming loop over 20,000 jobs, the tabular path costs a median of 0.076 ms per job and a 99th percentile of 0.188 ms; the relational network costs 0.434 ms per row plus 0.121 ms of graph lookup, and the GRU 0.576 ms. Charging the measured feature and inference latency to the predictive policies before they enqueue changes the closed gap by less than 0.001 at every load we ran, so the choice between these predictors is not a latency decision at this job size.

S2. Traces and Workload Characterisation

S2.1. The Field Audit of the Main Judging Trace

Section 7.2 of the manuscript states four readings of the main trace. This is the evidence for them, and the two archive defects that follow from reading the trace this way.

The timestamp is a result stamp.

Among later-period runs the gap to the same user's previous event grows one for one with the cost (slope 0.99 of the median gap on C) while the gap to the next event barely moves (slope 0.16), and the feedback after a user kills a run arrives within about 0.1 s whatever its length. A result stamp predicts both readings; a receipt stamp neither.

The whole-second leak.

Stamps are recorded to the whole second while costs are recorded to the microsecond, so the fractional part of a job's cost appears in the gap between its own stamp and the next event, and a feature like "seconds since the user's previous record" reads the label off the

clock. We close it by replacing t with $t + u$, where $u \in [0, 1)$ is drawn deterministically from a fixed key and the record's identifier.

The cost field and the limit.

The cost field is wall-clock occupancy of one evaluation including interpreter start-up, read from the 10th percentile of correct submissions (0.12–0.18 s across the two periods) and from its weak dependence on the number of test cases (median $0.196 + 0.007n$ s in the later period). The platform limit is 60 s: the last run to reach it ends at 2019-08-28 10:32, after which the largest observed cost is 59.97 s, with 13 runs in $[59, 60)$ s and none above. Before that date there was no fixed limit, the longest run is 8,901 s, and the 1,189 runs that end with a kill are almost all above 360.8 s, pointing at a container watchdog rather than judging demand.

Co-ending groups, zero-cost blocks and two short archives.

Groups of runs by the same user that end in the same second are 7.53% of blocks and 26.28% of capped work; every prediction table in the manuscript is repeated on rows outside such groups. A further 1,671 blocks record a cost of exactly zero with empty outputs; they leave the simulated trace and stay in the features, the labels and every total. The 2019-2 archive stops at 2019-09-19, months before that term's assessments end, and 2020-1 is short by about 3% of its blocks; both serve per-submission modelling and are excluded from anything that treats a term as a whole.

Against the published release statistics.

We match the class count in 18 of 18 terms and the student and exercise counts in 16 of 18, and parse 4.08% fewer execution blocks than the official column sums, whose printed totals disagree with those same columns by -248 exercises and $+246,246$ blocks. We quote the columns.

S2.2. *The Continuous-Integration Refit: Sharpness and Circularity*

Section 7.3 of the manuscript reports the held-out refit of the effective capacity k_{eff} and the setup term on the two Firefox CI pools. Two checks on that fit sit here.

The objective is sharp in k rather than flat: on a step-of-one grid only 5 to 8 values of k put the simulated mean wait within 10% of the recorded one, so the capacity the fit chooses is not one of many equally good answers.

The circularity of the original whole-window fit can be read off the two pools directly. On pool A the whole-window and held-out rows are almost identical, because every split picks nearly the same capacity; on pool B the reverse split's held-out capacity gives a mean ratio of 1.14 on its test part where the whole-window capacity gives 0.99. The difference between those two numbers is the size of the circularity that fitting on the reported window would have hidden.

What the split does not cover is stated in Section 9 of the manuscript and belongs here in full. Out of sample the 90th percentile lands up to 21% from the recorded one and is systematically low on the larger pool. The split is chronological inside one 21-day and one 7-day window of two pools, not a different cluster, quarter or workload, and the capacity is held constant over the test part although the real machine count varies.

S2.3. *What Supports the Shared-Pool Design*

This subsection supports the second and third of the three things Section 7.3 of the manuscript says support its shared-pool design: a measurement on the same demand, and the simulator reproducing waits somebody else recorded.

The second is a measurement on the same demand. Replaying the overlay of Section 7.4 of the manuscript under the dedicated design, the 99th percentile of containers alive at once is 1,636 and each container executes code for 0.12% of the time it is alive; since only 11% to 35% of logins have a logout written after them, session boundaries are reconstructed, and across six reconstructions that percentile lies between 1,003 and 2,525 containers and the utilisation between 0.07% and 0.17%. A shared pool holds the 99th percentile of deadline-window waits under 1 s with 14 servers under FCFS and 13 under the predictor-ordered policy. Pooling is worth two orders of magnitude here and the scheduler is worth one server (14 to 13 at 1 s, 11 to 10 at 5 s, 8 to 7 at 30 s). It does not support a claim that the platform ever queued, the pooled side being simulated, nor that the containers are over-provisioned, since they also run the editing session the pool does not replace, nor that any of it is a cost saving.

The third is that the simulator reproduces waits somebody else recorded. The CI logs carry the queue wait of every run, so we replay the recorded arrivals and services through our own k -server non-preemptive kernel. Two quantities are fitted: the effective capacity k_{eff} , the shortfall against the listed machine count being machines that are up but not claiming work, and a setup term, the median short gap between consecutive runs on the same machine, added to every service because a machine is not reusable the instant a run resolves. Fitting both on the window and then reporting agreement on that same window would be circular, so we refit both on one chronological part of each window and test on the other, five splits over the two pools. This split is internal to the recorded window and is held out of the fit only; it is not sealed in the sense of Section 7.6 of the manuscript, and the whole window has been read.

The capacity is stable across splits. On pool A the three splits choose 155, 155 and 157 of 173 listed machines, and on pool B the two splits choose 96 and 92 of 105; the setup term moves between 54.2 and 55.3 s on the first and between 126.2 and 128.6 s on the second. The objective is sharp in k . With those parameters frozen, the held-out part comes back at 0.90 to 1.14 times the recorded mean wait, 0.86 to 1.16 at the 99th percentile and 0.89 to 1.28 at the 99.9th, with a per-hour mean-wait correlation of 0.936 to 0.981, a per-job rank correlation of 0.940 to 0.963, and a median per-job absolute error of 16 to 84 s.

The 90th percentile does not survive the held-out test. It lands at 0.81 to 1.21 of the recorded one, and on the larger pool it is below the recorded value on all three splits while the 99th is above it, so the simulated distribution is too steep in its upper middle; the two-term objective hides this in sample by trading the mean against the p90, so we fit out of sample and say so. What the fit absorbs is nonetheless a property of the pool and no artefact of fitting: told instead that every listed machine is available, the same simulator underestimates the held-out mean wait by a factor of 1.3 to 2.6 and its per-hour correlation falls to 0.75. Section S2.2 gives the step-of-one sharpness check and reads the circularity of the original whole-window fit off the two pools directly.

S2.4. Cost Concentration and the Provenance of the Analysis Bound

This subsection carries the cost-concentration screen of Section 7.5 of the manuscript and says, trace by trace, what its arrival and service fields are and what each analysis truncation removes.

Arrival and service provenance across traces.

CodeBench's jittered result stamp minus the recorded duration is an execution-start proxy, not a measured receipt time. ACcoding has no submission clock: arrivals are drawn, including the Poisson comparison and the deadline-shaped construction; its submission ids determine ordering, not elapsed time. Its measured cost sums test case execution times,

Table S3. Share of executed work carried by the longest jobs, and provenance of the analysis bound L . An analysis cap is applied to the replay, not evidence of a platform-enforced timeout.

Trace	Top 1%	Top 5%	Bound and provenance
CodeBench, 60 s regime	0.375	0.432	60 s, observed regime ceiling; analysis cap
CodeBench, no-limit periods, raw	0.941	0.960	none
CodeBench, no-limit periods, capped	0.494	0.658	60 s, imposed analysis truncation
ACcoding, measured cost	0.275	0.628	30.0 s, observed maximum of measured test-cost sum
ACcoding, with the assumed fixed term	0.122	0.290	30.5 s, observed maximum after assumed overhead
Serverless invocations	0.443	0.879	286.8 s, training p99.9 analysis cap
Compute farm	0.372	0.628	493,857 s, training p99.9 analysis cap
CI pool A	0.034	0.107	5,066 s, training p99.9 of run plus setup; clipped
CI pool B	0.052	0.159	4,140 s, training p99.9 of run plus setup; clipped

each subject to its problem limit, and the replay adds an assumed 0.5 s overhead. Thus neither the observed 30.0 s sum nor the resulting 30.5 s maximum is a verified whole-job timeout.

For Azure, the replay arrival is the documented end timestamp minus duration, again a start proxy; the publisher explicitly states that invocation timestamps were modified from the production trace [2]. The Netbatch replay uses the recorded SWF submission time and run duration. Their bounds in Table S3 are training p99.9 truncations, not observed platform timeouts. Both FCFS and every reordered policy receive the same truncated service sequence. The truncation affects 1,187 Azure jobs and 7,696 Netbatch jobs in the analysed pools; it removes 3.18% and 2.16% of raw work.

In the CI pools, arrival is the recorded scheduled instant, when dependencies have cleared and a worker may claim the run; recorded wait is started minus scheduled. Service is resolved minus started plus a fitted same-worker setup term. The analysis first adds setup and then clips this occupancy at the training p99.9 L ; longer jobs are retained with truncated service, not excluded. This clips 227 runs in pool A and 27 in pool B. FCFS and all comparison policies use exactly that same capped occupancy sequence. The source scheduler's priority-then-FIFO rule and operational capacity are distinct from the counterfactual FCFS model used for the guarantee.

An enforced L in a deployed service would instead kill an unfinished job at the limit and charge that executed work to the same-input FCFS comparison. The killed job would not yield its intended result; these replays do not estimate the resulting completion-quality cost or users' resubmission response. The table therefore supports finite-input bounds under its stated transformations, not a future operational timeout claim.

S3. Experiments

S3.1. The Experimental Configuration

This subsection carries the configuration and selection-protocol table that Section 8.1 of the manuscript refers to, unchanged and under its own label.

S3.2. What the Selection Rule Chose

Table S4 gives the three pre-stated grids, the feasibility rule, the objective and the tie-break, and names the shape the rule selected at each promise. Table S5 gives

Table S4. Experimental configuration and the selection protocol for the guard parameters.

Item	Value
Trace and pool	CodeBench development terms under the 60 s limit, overlaid as in Section 7.4 of the manuscript; 17,634,760 jobs, 30 weeks, five overlays
Load levels	$k = 8/5/4$ on four of the five overlays and $7/5/4$ on the fifth, whose work per busy hour is the lowest; busy-hour $\rho = 0.529/0.847/1.059$ on overlay 0
Single-server configuration	separate trace, 321 copies, 3,209,347 jobs, busy-hour $\rho = 0.80$ at $k = 1$
Primary metric	99th percentile of the wait among jobs arriving in the 24 h before their own deadline; secondary metrics as defined in Section 3 of the manuscript, the gap closed printed beside the percentage reduction
Intervals	paired week-block bootstrap, 2,000 resamples of the 30 whole weeks, one draw of week multiplicities applied to every policy and to all five overlays
Promise to parameter	$B_{\max} = k(G - (3 - 2/k)L)$ with $L = 60$ s, no data used
Budget shapes	one mechanism under one cap, three shapes: constant ($\gamma = \eta = 0$), age-relative ($+\eta k(t - a_q)$), queue-length ($+\gamma n_q$). Three pre-stated grids, $42 + 162 + 54 = 258$ points
Selection rule	a point is feasible if its harm is $\leq G/2$ at each of the three load levels of each of the five validation overlays (15 cells); the objective is the deadline-window gap closed in the worst cell; ties to smaller worst harm, then smaller B_0 , then η , then γ . Guard(G) is the winner over the three grids together; the winner inside each grid is an ablation row
What the rule selected	Age-relative at $G = 300$ and 600 s ($B_0 = 60$ and $120k/4$ s, $\eta = 0.50$ and 0.75), queue-length at $G = 1200$ s ($\gamma = 16k/4$ s). Table S5 gives the three selected points with their worst-cell gap and harm, the feasible counts and the nine per-shape winners; the same values are in <code>protocol_lock.draft.json</code>
Comparators	As named in Table 1 of the manuscript. Fixed(G) is the equal-promise budget $B_{\max}$, which the harm rule rejects at every promise; Skip(G) takes the largest dispatch-charged count equation (2) of the manuscript allows at that G and that cell's k

the selected points in full and the nine per-shape winners. The same values are in `protocol_lock.draft.json`, which is the artefact the selection was frozen into.

Table S5. The three selected budget points and the nine per-shape grid winners. Budgets in work scale as $k/4$; the harm limit is $G/2$, the feasibility threshold of the selection rule. “Feasible” counts the grid points that met it in all 15 validation cells.

Item	Value
Selected, $G = 300$ s	age-relative: $B_0 = 60k/4$ s, $\eta = 0.50$, $\gamma = 0$; worst-cell gap 0.4251, worst-cell harm 122.8 s against a 150 s limit, 44 of 99 candidate points feasible
Selected, $G = 600$ s	age-relative: $B_0 = 120k/4$ s, $\eta = 0.75$, $\gamma = 0$; worst-cell gap 0.7216, worst-cell harm 293.3 s against 300 s, 70 of 107 feasible
Selected, $G = 1200$ s	queue-length: $B_0 = 0$, $\eta = 0$, $\gamma = 16k/4$ s; worst-cell gap 0.7283, worst-cell harm 145.3 s against 600 s, 95 of 111 feasible
Per-shape winners	constant: $B_0 = 329.671, 981.126, 2,029.958 k/4$ s at the three promises; age-relative: $(60, 0.50), (120, 0.75), (360, 0.75)$; queue-length: $\gamma = 16k/4$ s with $B_0 = 0, 120k/4$ s, 0

S3.3. The Two Budget Shapes the Main Table No Longer Prints

Section 8.5 of the manuscript compares the three budget shapes at a fixed promise through the paired differences, which is how the comparison is read. Table S6 gives the age-relative and queue-length rows themselves, in the format of Table 5 and in the same cells; the Guard(G), Guard-fixed(G), Fixed(G) and Skip(G) rows stay there. One of the three shape winners coincides with Guard(G) at each promise, which is why some rows below repeat a row of Table 5.

Table S6. The age-relative and queue-length winners at three promises and three loads, in the format of Table 5 of the manuscript. Excess and harm are the worst over the five overlays, the rest are means; budgets scale as $k/4$.

Load	Policy	Promise	p99 _{dl}	Gap closed	Red. (%)	Max exc.	Harm	Fired (%)
$\rho = 0.5, k = 7.8$								
	Guard-age(300)	300	15.58	0.705 [0.678, 0.724]	43.7	192.5	132.1	1.35
	Guard-queue(300)	300	15.27	0.723 [0.697, 0.742]	44.8	193.4	94.3	21.60
	Guard-age(600)	600	14.97	0.740 [0.713, 0.760]	45.8	492.2	237.0	0.17
	Guard-queue(600)	600	14.88	0.744 [0.718, 0.765]	46.1	504.5	106.0	0.16
	Guard-age(1200)	1200	14.73	0.751 [0.723, 0.773]	46.5	1,049.1	237.0	0.01
	Guard-queue(1200)	1200	14.84	0.746 [0.719, 0.768]	46.2	907.8	94.3	22.01
$\rho = 0.8, k = 5$								
	Guard-age(300)	300	55.94	0.738 [0.697, 0.761]	53.0	209.7	114.1	11.57
	Guard-queue(300)	300	54.69	0.753 [0.713, 0.777]	54.0	210.7	117.6	17.53
	Guard-age(600)	600	47.47	0.837 [0.815, 0.851]	60.1	507.8	319.6	1.31
	Guard-queue(600)	600	47.69	0.834 [0.814, 0.847]	59.9	508.5	117.7	1.35
	Guard-age(1200)	1200	45.16	0.864 [0.844, 0.876]	62.0	1,102.9	370.5	0.29
	Guard-queue(1200)	1200	46.85	0.844 [0.823, 0.857]	60.6	1,104.9	117.6	9.93
$\rho = 1.0, k = 4$								
	Guard-age(300)	300	163.40	0.478 [0.387, 0.602]	40.2	229.1	134.8	28.64
	Guard-queue(300)	300	169.22	0.453 [0.368, 0.607]	38.1	231.4	144.8	30.99
	Guard-age(600)	600	89.24	0.801 [0.764, 0.842]	67.3	523.7	339.9	7.15
	Guard-queue(600)	600	91.09	0.793 [0.758, 0.833]	66.6	515.8	123.4	7.08
	Guard-age(1200)	1200	72.52	0.874 [0.851, 0.893]	73.4	1,117.3	453.3	0.81
	Guard-queue(1200)	1200	82.67	0.830 [0.801, 0.857]	69.7	1,110.4	144.8	6.21

S3.4. Deliberately Corrupted Scores

This subsection carries the table behind the manuscript’s answer in Section 8.5 to what happens when the predictions are wrong: the guard holds every job inside its promise under a reversed, a random and a longest-called-shortest score, while the deadline-window percentile is not rescued.

Table S7. Deliberately corrupted scores, unguarded and under Guard(600). Waits and excesses in seconds; corruption of the original source-clock scores.

Load	Score	p99 _{dl}		Max excess		Harm guarded
		unguarded	guarded	unguarded	guarded	
$\rho = 0.5$	reversed	42.72	43.39	1,240.1	497.1	287.1
	random	16.82	17.12	1,307.7	489.9	324.0
	longest as shortest	38.24	38.71	1,305.8	497.1	273.4
$\rho = 0.8$	reversed	228.12	232.41	3,387.9	513.5	323.4
	random	78.52	88.39	3,319.4	504.5	352.3
	longest as shortest	212.54	222.52	3,292.0	516.1	306.0
$\rho = 1.0$	reversed	557.44	546.48	6,295.7	534.8	309.1
	random	194.78	261.93	6,235.5	520.7	332.9
	longest as shortest	523.55	543.63	6,107.8	534.7	324.6

S3.5. The Residual on All Nine Cells

Table 6 of the manuscript prints the three Guard(600) cells and one reversed score. Table S8 gives all nine. The reading Section 8.5 takes from them is that the residual is a property of the schedule and barely moves with the policy: inside each load the ratios are similar in scale, but are not equal to the third decimal. An R^2 rounded to unity is likewise not exact equality.

Table S8. The residual $D_i = k(W_P[i] - W_{FCFS}[i]) - (\text{In}_i - \text{Out}_i)$ of Theorem 1 of the manuscript on overlay 0, 17,634,760 jobs per cell. Ratio is $\max |D_i|$ over the bound $2(k - 1)L$, which is 14L, 8L and 6L at the three loads; R^2 and the worst error are for $(\text{In}_i - \text{Out}_i)/k$ against the realised excess over first-come first-served.

Load	Policy	$\max D_i /L$	Ratio	$D_i = 0$	R^2	worst err. (s)
$\rho = 0.5, k = 8$	SPJF-E	6.788	0.485	95.6%	0.985	50.9
	Guard(600)	6.788	0.485	95.6%	0.983	50.9
	Reversed score	6.788	0.485	95.6%	0.996	50.9
$\rho = 0.8, k = 5$	SPJF-E	3.922	0.490	87.2%	0.996	47.1
	Guard(600)	3.922	0.490	87.2%	0.995	47.1
	Reversed score	3.881	0.485	87.5%	0.999	46.6
$\rho = 1.0, k = 4$	SPJF-E	3.082	0.514	81.1%	0.998	46.2
	Guard(600)	3.082	0.514	81.1%	0.998	46.2
	Reversed score	2.994	0.499	81.7%	1.000	44.9

S3.6. A Single Server

Section 8.6 of the manuscript reports the single-server configuration through five gap figures, three harm ratios and two realised excesses. Table S9 is the configuration in full. The parameters are the ones the selection rule chose on the 15 multi-server validation cells; $k = 1$ is not among them, and we do not re-tune on it.

The firing rate of the three unguarded policies is printed ---: the column counts dispatch epochs at which the guard overrides the base policy, and these three runs have no guard. Scoring first-come first-served at 100% by the convention that every first-come first-served dispatch serves the queue head, as an earlier version of this table did, compares it with the guarded runs on a quantity it does not have.

Table S9. Single server, busy-hour $\rho = 0.80$, 3,209,347 jobs. Waits and excesses in seconds. The parameters are the ones the rule selected on the multi-server validation cells; this configuration is not among them. The skip count is $\kappa = \lfloor Gk/L \rfloor - (2k - 2)$.

Policy	Promise	p99 _{dl}	Gap closed	Max exc.	Harm	Fired (%)
FCFS	—	221.27	0.000	0.0	0.0	—
SJF on true sizes	—	45.32	1.000	2,054.5	777.9	—
SPJF-E	—	60.91	0.911 [0.854, 0.933]	2,520.3	656.4	—
Guard(300)	300	89.96	0.746 [0.571, 0.829]	297.9	81.8	8.17
Guard-fixed(300)	300	125.68	0.543 [0.207, 0.718]	140.3	119.5	17.09
Guard-age(300)	300	89.96	0.746 [0.571, 0.829]	297.9	81.8	8.17
Guard-queue(300)	300	105.15	0.660 [0.532, 0.715]	297.9	53.9	29.15
Fixed(300)	300	83.18	0.785 [0.584, 0.865]	297.9	246.3	6.33
Skip(300), $\kappa = 5$	300	220.39	0.005 [-0.002, 0.019]	95.1	81.8	73.17
Guard(600)	600	71.07	0.854 [0.780, 0.895]	595.5	165.5	2.53
Guard-fixed(600)	600	82.10	0.791 [0.602, 0.868]	303.0	246.3	6.16
Guard-age(600)	600	71.07	0.854 [0.780, 0.895]	595.5	165.5	2.53
Guard-queue(600)	600	91.52	0.737 [0.640, 0.812]	582.0	81.8	3.43
Fixed(600)	600	68.48	0.868 [0.783, 0.906]	593.4	563.6	2.29
Skip(600), $\kappa = 10$	600	219.89	0.008 [-0.004, 0.037]	100.9	83.2	66.98
Guard(1200)	1200	98.00	0.701 [0.640, 0.742]	1,157.4	53.9	26.15
Guard-fixed(1200)	1200	69.03	0.865 [0.789, 0.905]	564.2	509.3	2.57
Guard-age(1200)	1200	63.80	0.895 [0.835, 0.920]	1,182.3	366.4	0.68
Guard-queue(1200)	1200	98.00	0.701 [0.640, 0.742]	1,157.4	53.9	26.15
Fixed(1200)	1200	63.19	0.898 [0.837, 0.923]	1,180.4	656.4	0.45
Skip(1200), $\kappa = 20$	1200	217.64	0.021 [-0.008, 0.084]	168.0	97.7	58.67

S3.7. Outcome Visibility: Exposure, Lag and the Variants

This subsection supports the conservative-visibility headline of Section 8.3 of the manuscript: it defines the same-copy queueing exposure the headline conditions on, audits the release lag, and gives the two controls and the pending exact variant that the manuscript's headline table names.

Let H_i be the set of distinct outcome records used by score i , counting a record only once if it enters several entity aggregates. Separate records from the same class-term copy, histories from other classes or terms, and records with no submission job in the replayed pool. Same-copy queueing exposure is

$$V_i(P) = \{j \in H_i : \text{copy}(j) = \text{copy}(i), \quad a_j + W_P[j] + C_j > a_i\}.$$

An external past record is not unavailable merely because it has no replayed job. The earlier audit mapped histories through the same replication round while different classes were independently shifted. Its roughly 80% affected jobs and 95.39% with unreplayed history are mapping-exposure statistics, not estimates of same-copy queueing leakage. Section S5 specifies the external-history clock and the separate class-history sensitivity.

The conservative score restricts user, exercise and pair history to submission outcomes in the same class-term copy, and admits a result only when $\text{done}_j + 3600 \leq a_i$, with times in seconds. The static score uses code, course and clock information known at submission, without outcome history. Both use their own fitted models; original, conservative and static therefore differ in more than release delay. Their differences cannot be attributed to queueing delay alone and are not mathematical upper or lower performance bounds. Guard parameters remain those selected under the original score. Consistent with the committed visibility decision, the conservative variant is the empirical headline; the original-score results are optimistic references.

For a same-copy submission, $W_P[j] \leq D$ is a sufficient availability condition under a source release lag D . The largest development FCFS wait is 1544.602708 s. Adding the

largest evaluated promise, 1200 s, still lies below the 3600 s lag. All conservative Guard and FCFS cells have 0 lag violations. Unguarded conservative SPJF-E has 336 job occurrences with waits beyond the lag across the evaluated cells; it lacks the guard’s certificate, so its performance is a lagged-score comparator with residual availability violations. A new pool must pass the per-job lag audit; the development maximum does not certify it in advance.

The exact variant freezes the original model weights, replays each policy, withholds same-copy outcomes still unavailable at the target arrival, recomputes affected histories and scores, and repeats until no used outcome violates availability. Withheld sets only grow, so this finite procedure terminates. It certifies availability in its final replay; it is not a claim of minimal withholding, a unique fixed point, or an online predictor driven by completion events. A full comparison and the associated exposure decomposition remain pending. The conservative and static results above are the completed controls; no partial exact result is used to select parameters or replace the headline.

S3.8. Aging Outside the Budget Family

This subsection supports the manuscript’s statement in Section 8.3 that a linear aging credit, selected under the same conservative score and the same harm limit, carries no proved per-job guarantee and closes far less of the gap.

Table S10 extends the policy definitions with linear aging. It uses the same conservative score as the headline, with a validation search over 11 credits, of which 4 pass the 300 s harm limit. The selected credit is $\beta = 0.03$; its worst validation gap is 0.0375 and harm 246.36 s. The label Aging(600) records the reference promise used to set the validation constraint, and carries no proved per-job guarantee. All evaluated loads and harmful outcomes are retained.

Table S10. Additional policy definition; the reference label is not a theorem.

Policy	Dispatch priority, smallest first	Guarantee
Aging(G)	$\hat{C}_i - \beta(t - a_i)$, equivalently $\hat{C}_i + \beta a_i$; ties by arrival rank. Select β on validation cells under harm $\leq G/2$.	None proved

Table S11. Conservative-score comparison with the selected aging baseline. All waits, excesses and harms are in seconds; intervals are paired week-block intervals.

Load	Policy	p99 _{dl}	Gap closed	Max excess	Harm
0.5	Guard(600)	16.68	0.640 [0.604, 0.671]	499.9	235.2
0.5	Aging(600)	23.83	0.228 [0.180, 0.271]	306.8	155.8
0.8	Guard(600)	64.77	0.634 [0.560, 0.688]	509.4	327.7
0.8	Aging(600)	109.83	0.111 [0.082, 0.161]	673.0	179.1
1.0	Guard(600)	174.71	0.429 [0.357, 0.533]	527.8	303.5
1.0	Aging(600)	261.68	0.051 [0.035, 0.074]	855.6	184.7

S3.9. Package Predictor Metrics and Their Splits

This subsection supports the manuscript’s distinction in Section 8.3 between the AUROC of the earlier retrospective classifier comparison and the AUROC of the deployed expected-cost score.

The AUROC 0.928 in the earlier predictor comparison is M4 under the source-clock I-res protocol, using heavy-class classifier probabilities averaged over seeds, trained on the core terms and evaluated on 2022-2. It is not the AUROC of the package’s deployed expected-cost score. The latter is 0.9360 pooled over rolling forward development predictions and 0.9372 on 2022-2; the conservative pooled score is 0.9116. Table S12 uses

the package values and split definitions that will accompany the sealed predictor column. The earlier value remains evidence for its own retrospective classifier comparison, whose predictions never enter scheduling; it is not a competing estimate of the same fitted score.

Table S12. Package development AUROC with user-block intervals. Pooled means pooling unrepliated forward-prediction rows, not averaging term AUROCs.

Target	Score variant	AUROC [interval]
pooled	Original	0.9360 [0.9289, 0.9424]
pooled	Conservative	0.9116 [0.9054, 0.9173]
pooled	Static	0.8371 [0.8249, 0.8492]
2022-2	Original	0.9372 [0.9218, 0.9491]
2022-2	Conservative	0.9196 [0.9060, 0.9324]
2022-2	Static	0.8806 [0.8544, 0.9022]

S3.10. *Alternative Ranking Losses*

This subsection supports the manuscript’s answer in Section 8.4 to the question of what a scheduler should rank by, by running seven further fitted scores through the same simulator.

Alternative ranking losses were fitted on the same features and run through the same simulator (Table S13). The fitted quantile scores and raw-second squared-error model lose to the Tweedie fit at every load in this comparison. Raw squared error and the Tweedie mean loss have the same unrestricted population target, so this does not identify a change of estimand as the cause. Finite model capacity, error weighting, regularisation and optimisation can produce different rankings for the same target. A two-class rule that sends predicted-heavy jobs behind predicted-light ones and runs FCFS inside each class reaches a deadline-window p99 of 64.05 s at $\rho = 0.8$ on the first overlay, against 43.86 s for the full ordering on the same overlay, so most of the benefit comes from the fine ordering inside the light class and little of it from separating the two classes.

Table S13. Ranking scores fitted on one feature set and run through one simulator: fraction of the deadline-window gap closed, five overlays, and the paired difference against the log-scale score. Archived source-clock fits, distinct from the package refit in Table 4 of the manuscript; differences between the tables do not compare identical fitted scores.

Score	$\rho = 0.5$	$\rho = 0.8$	$\rho = 1.0$	Difference at $\rho = 1.0$
SPJF-log	0.7528	0.8728	0.8967	—
SPJF-E	0.7515	0.8847	0.9159	+0.0191 [+0.0111, +0.0238]
Expected cost, gamma hurdle	0.7404	0.8826	0.9141	+0.0174 [+0.0082, +0.0223]
Probability of heavy	0.7476	0.8821	0.9122	+0.0155 [+0.0066, +0.0199]
Predicted 99th percentile	0.7245	0.8742	0.9005	+0.0038 [−0.0050, +0.0097]
Predicted 90th percentile	0.7024	0.8470	0.8755	−0.0212 [−0.0297, −0.0156]
Predicted median	0.7071	0.8196	0.8402	−0.0565 [−0.0684, −0.0463]
Squared error, raw seconds	0.7197	0.8585	0.8873	−0.0094 [−0.0194, −0.0043]

S3.11. *What the Shape of the Budget Is Worth*

This subsection supports the manuscript’s statement in Section 8.5 that the cap carries the promise and the shape decides only how much of the base policy’s benefit survives and which jobs pay for it.

The promise is the cap’s alone, so the three shapes are compared at a fixed promise and the question is only how much of the base policy’s benefit each keeps and who pays for it. The two shapes the main table no longer prints are read here through the paired differences and in full in Table S6. The shape matters where the promise is tight and the

load is high, and hardly at all elsewhere. At $G = 600$ s the constant shape trails the joint winner by 0.007 [0.003, 0.012], 0.028 [0.020, 0.038] and 0.119 [0.074, 0.176] of the gap as the load rises through the three levels, each interval clear of zero; at $G = 300$ s the same three differences are 0.008 [-0.002, 0.015], 0.099 [0.056, 0.157] and 0.262 [0.202, 0.288]; and at $G = 1200$ s they are -0.001 , -0.001 and 0.003, none of the three intervals excluding zero. At a loose promise and a light load the shape is not doing measurable work.

The other axis is harm. At $G = 600$ s the queue-length shape reaches within 0.009 of the joint winner's gap at every load while its worst-overlay harm is 106.0, 117.7 and 123.4 s against the winner's 237.0, 319.6 and 339.9 s, under half at every load. That is also why the rule picks the queue-length shape at $G = 1200$ s, where the three shapes are within 0.005 of one another on the validation objective and it has the smallest worst harm. The shape decides which jobs pay for the promise more than it decides how much of the benefit survives.

Fixed(G), which spends the whole cap, is an equal-promise candidate in the search, but is infeasible under the validation harm constraint at every promise. Its evaluation-overlay harm is 181.8, 195.4 and 208.2 s at $G = 300$ s against a 150 s limit, 237.0, 447.1 and 465.5 s at $G = 600$ s against 300 s, and 237.0, 469.2 and 708.8 s at $G = 1200$ s against 600 s. Thus the evaluation cells also exceed the corresponding limits at each promise. It is ahead of the joint winner on the gap closed in eight of the nine cells, by up to 0.042, so it buys its extra gap with harm above the declared limit in some cells; we make no claim that any shape beats it on the gap at equal promise.

S3.12. *The Gap Closed against the Promise*

This subsection carries the figure behind the manuscript's Section 8.5: the fraction of the gap each policy closes, plotted against the promise, with the count-based baseline beside it.

S3.13. *The Identity, Measured Job by Job*

Section 6 of the manuscript states that neither exhaustive search nor real input reaches the error term of Theorem 1. Exhaustive and random search over millions of small instances never exceeds $(k - 1)L$ by more than one time unit, and on the 17.6-million-job trace of Section 8 of the manuscript the largest residual reaches 0.470 to 0.518 of $2(k - 1)L$, again about $(k - 1)L$.

This subsection supports the manuscript's report in Section 8.6 that the residual of Theorem 1 never exceeds 0.518 of its bound on the trace, by giving the recording kernels, their validation and the same-phase shares behind that figure.

The net-overtake identity had been tested on small synthetic instances and never measured on the trace this paper runs on, so we measured it. Table 6 of the manuscript reports the residual $D_i = k(W_P[i] - W_{FCFS}[i]) - (In_i - Out_i)$ on overlay 0 at the three load levels under Guard(600), with the reversed score at the busiest load beside it and the other five cells, SPJF-E among them, in Table S8, and with FCFS on the same input as the reference schedule in every cell. Recovering In_i and Out_i needs the dispatch sequence, which the guard kernel does not expose, so the kernel and the independent reference simulator were each copied verbatim and given recorders that add no branch and no state update; the two agree on the dispatch sequence of every job of 3,000 random instances, the recording kernel reproduces the production kernel's waits element for element inside every cell, and In_i and Out_i are carried in exact integer microseconds, the currency the guard itself meters.

Theorem 1 of the manuscript is asserted on every job of every cell: 0 violations over 1,428,415,560 job-checks, the 27 policies of this overlay at the three loads. The residual stays far from its ceiling. The largest $|D_i|$ over the nine cells of Table S8 is $6.788L$, at $k = 8$,

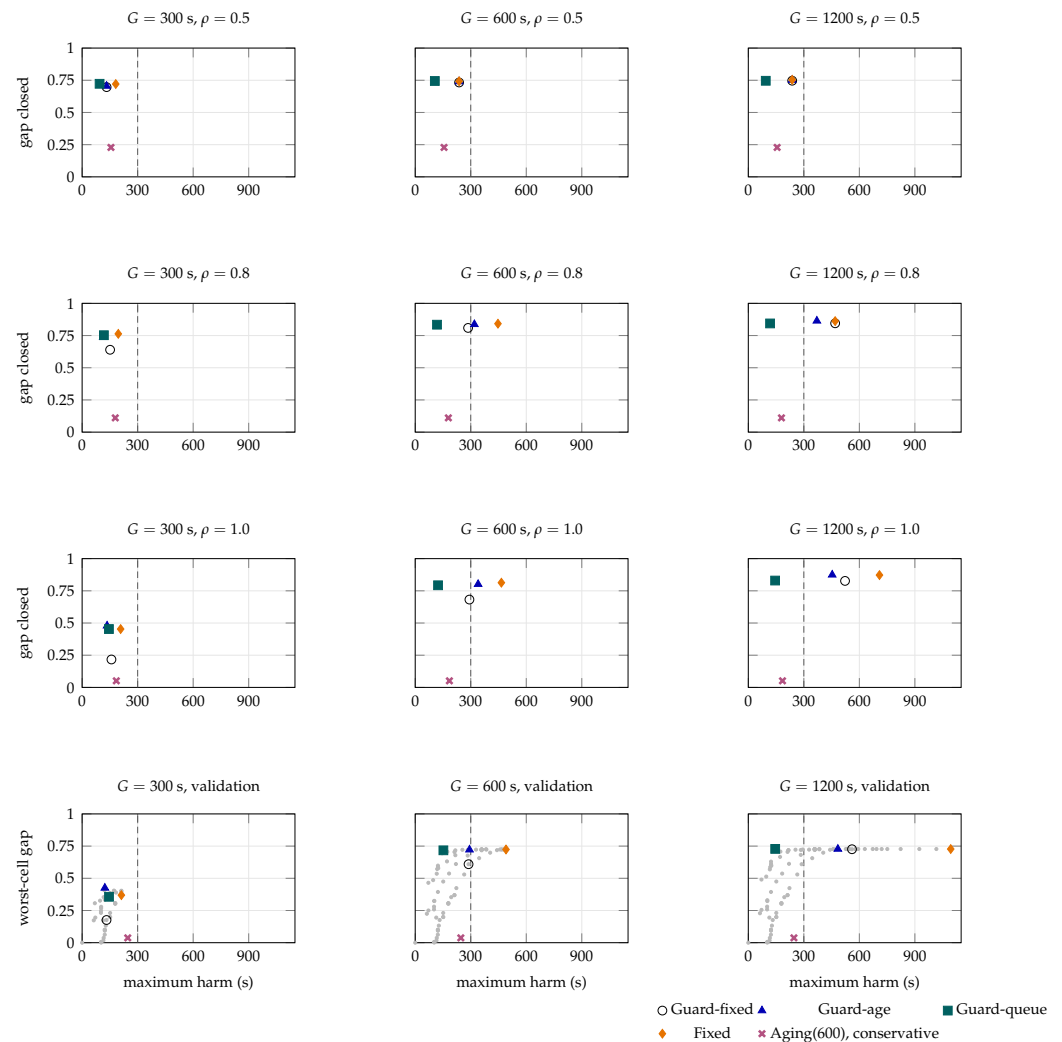


Figure S1. Gap-harm trade-offs for every selected budget shape and Fixed(G), at each promise and load (upper rows); validation candidates and per-family selections (bottom row). A joint winner coincides with its selected family point. The dashed line is the 300 s validation limit for $G = 600$ s; it is shown in every panel as a common reference, not as the constraint for every promise. The other promise-specific limits follow $\text{harm} \leq G/2$. All budget-family points use original scores. The magenta point is the only selected aging comparator, Aging(600), with conservative scores and no proved guarantee; it is repeated across promise panels for orientation, not returned at other promises. Cross-protocol points do not establish Pareto dominance. Maxima have no population interval; gap intervals for the selected rows are reported in the tables.

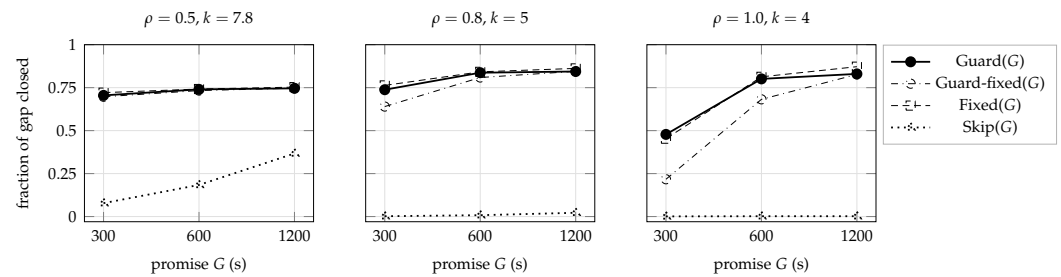


Figure S2. Original source-clock scores (optimistic reference): what each budget shape keeps of the FCFS-to-SJF gap as the promised excess G is relaxed. Every point is a “Gap closed” cell of Table 5 of the manuscript; the intervals are in that table and are omitted here. All four policies carry the same promise at a given G , since the promise depends on the cap alone. The gap between Guard(G) and its constant-shape ablation opens where the promise is tight and the load is high and closes elsewhere; counting positions rather than work (Skip) collapses at the two higher loads; Fixed(G), which spends the whole cap, buys its gap with harm the selection rule rejects.

where the bound $2(k-1)L$ is $14L$, and over those nine cells the maximum sits between 0.485 and 0.514 of the bound, which is about $(k-1)L$; over all 81 cells the largest is 0.5173 of the bound, so it never exceeds 0.518. That is where a search that does not build the cascade of Theorem 2 of the manuscript stops, so the measurement says the workload does not attain the constant, not that the constant is loose. Same-phase bookkeeping, the mechanism behind the degenerate extremal instances of Remark 1 above, accounts for at most 0.36% of the total $\ln$ in any of those nine cells, so the executed-work convention would not change the picture for these policies. It is not negligible everywhere: under the randomised ranking, which ties many jobs into one dispatch instant, the share reaches 14.95%. $(\ln_i - \text{Out}_i)/k$ explains $R^2 = 0.983$ to 0.9996 of the variance of $W_P[i] - W_{\text{FCFS}}[i]$, with a worst per-job error of 44.9 to 50.9 s where the theorem permits $(2 - 2/k)L$, that is 90 s at $k = 4$ and 105 s at $k = 8$. This is one overlay of development data, and the residual is a property of a schedule rather than a statistic, so no interval is attached to it.

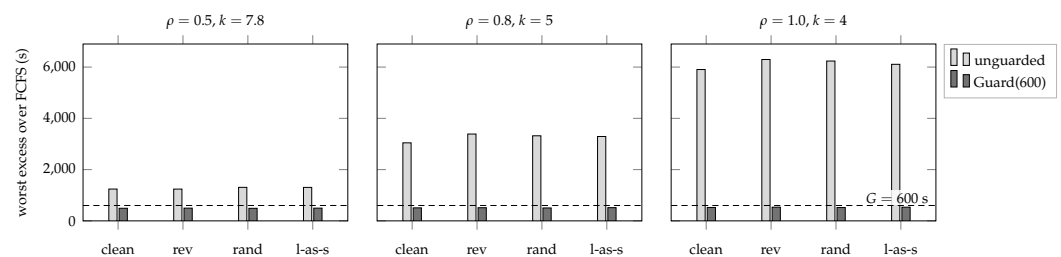


Figure S3. Original-score reference and its corrupted-score controls. The worst single job, with and without the guard, under the expected-cost score (“clean”) and the three corrupted scores of Table S7: reversed (rev), random (rand) and truly-longest-called-shortest (l-as-s). The dashed line is the promise $G = 600$ s. Sources: the clean pair is the block header and the Guard(600) row of Table 5 of the manuscript; the three corrupted pairs are the “Max excess” columns of Table S7. The unguarded bar grows with load and with corruption; the guarded bar does not, which is the content of Theorem 3 of the manuscript and not a property of the trace.

S3.14. Choosing a Shape on One Set of Cells

This subsection supports the manuscript’s statement in Section 8.7 that the budget shape selected on the fifteen multi-server validation cells does not transfer to a single server at the loosest promise.

The single-server configuration also shows the cost of choosing a shape on one set of cells. The parameters were selected on the 15 multi-server validation cells, and $k = 1$ is

not among them. At $G = 300$ and 600 s the joint winner transfers: it closes 0.746 and 0.854 of the gap, against 0.543 and 0.791 for the constant shape. At $G = 1200$ s it does not: the queue-length shape that wins the joint comparison on the multi-server cells closes 0.701 here, while the age-relative shape closes 0.895 and the constant shape 0.865 . We report the number the pre-stated rule produces and do not re-tune it on this cell; Section 9 of the manuscript states what it means.

S3.15. Trace-Construction Sensitivities

This subsection supports the manuscript’s statement in Section 8.10 that the earlier ordering experiment survives five alternative readings of the source log, and that those variants were not rerun under the final selected policy.

Five trace-construction variants were run through an earlier source-clock pipeline (Table S14). These are sensitivity results for its stored M4 log-scale score and fixed completed-work budget, not the selected expected-cost Guard(G). Within that earlier configuration, the alternatives retain a positive gap; adding the term whose busiest hour is dominated by runs that outlived their container raises the gap, and the opposite reading of the timestamp, in which the stamp is receipt rather than completion, gives 0.926 against 0.895 . Merging or dropping the groups of runs by one user that end in the same second moves the gap closed by 0.017 and 0.003 at full load, so that earlier result survives these two treatments of co-ending runs; delaying the visibility of a completed job’s outcome costs prediction quality and not much of it (Section 4 of the manuscript). The earlier fixed-budget guard is more sensitive to the construction, ranging from 0.442 to 0.843 of the gap at a fixed budget of 600 s at full load.

Table S14. Trace-construction sensitivities at busy-hour $\rho = 1.0$, three overlays, fraction of the deadline-window gap closed. Archived M4 log-scale score and fixed completed-work budget; the budget in the last column is not a promise of equal value.

Variant	k	FCFS $p99_{dl}$	Unguarded	Guard, budget 600 s
Primary pool (overlays 0–2)	4.000	—	0.895 [0.864, 0.916]	0.442 [0.345, 0.631]
Earlier term added	4.667	95.41	0.945 [0.885, 0.965]	0.843 [0.715, 0.880]
Co-ending runs merged	4.000	228.19	0.912 [0.892, 0.924]	0.603 [0.445, 0.740]
Co-ending runs dropped	4.000	209.30	0.898 [0.879, 0.914]	0.608 [0.471, 0.753]
Opposite timestamp reading	4.000	268.75	0.926 [0.904, 0.939]	0.545 [0.422, 0.699]

Table S14 supports sensitivity of the earlier ordering experiment to timestamp interpretation, source-term inclusion and co-ending-run treatment. It does not validate robustness of the final selected Guard(G), conservative history or policy-specific exact visibility. Those variants were not rerun under each altered trace construction. Section S5 gives the configuration and release boundary explicitly.

S3.16. Further Measurements from the Guarded Runs

This subsection carries three measurements that Section 8 of the manuscript states in summary: the worst wait among heavy jobs, how the realised worst excess compares with the bound at each k , and what the accounting of the count baseline is worth.

The worst wait among heavy jobs, over the five overlays, is 487 , $1,023$ and $1,543$ s under FCFS at the three loads, $1,375$, $3,313$ and $6,286$ s under SPJF-E, and 921 , $1,467$ and $1,993$ s under Guard(600). The guard therefore leaves the worst heavy job within 434 , 444 and 450 s of the FCFS figure, inside the promise, where the unguarded ranking multiplies it by 2.8 , 3.2 and 4.1 .

The ratio of the realised worst excess to the bound reaches 0.9950 at $k = 1$, 0.9311 at $k = 4$, 0.9219 at $k = 5$, 0.8975 at $k = 7$ and 0.8850 at $k = 8$. The finite-skip bound, by contrast, is attained exactly on small instances, at a ratio of 1.0000.

Replacing the largest 1% of underestimates by the true cost moves the deadline-window p99 at $\rho = 1.0$ from 66.57 s to 45.01 s on the first overlay and from 65.02 s to 45.32 s on the second; repairing the largest 1% of overestimates instead moves it to 60.37 s and 59.72 s. Section 8.4 of the manuscript prints the four fractions of the distance to the SJF reference that these give, and the two at 0.1%.

The count baseline is charged at dispatch, which allows 34 rather than 30 passes at $k = 4$ and $G = 600$ s; charging at completion instead changes the gap closed by at most 0.015, and by at most 0.001 at the two higher loads.

S4. Deferred Proofs

Section 6 of the manuscript gives the ideas behind the arguments proved here. The workload comparison uses continuity and left limits, whereas consistency is proved by induction; the strictness at $k \geq 2$ runs the same way as the bound itself, in that at the first instant t_1 at which $D = U_{Q_1} - U_{Q_2}$ touches $(k-1)L$ the linear piece to its left must have positive slope, which forces at most $k-1$ jobs present under Q_1 there, so none of them waits, and left limits then make U_{Q_1} strictly decreasing on that piece, so it cannot equal its own supremum at t_1 . The consistency induction needs two further readings, both one line: its first consequence uses $\text{over}[q](t) \leq \text{In}_q^A$, a completed overtaker of q having been dispatched while q waited, and the second uses Corollary 2 of the manuscript. The tightness constructions and the remaining proofs are Section S8 of this file. We use the manuscript's notation: $U_Q(t)$ is the unfinished work at time t under policy Q , and $N_Q(t)$ is the number of jobs present.

S4.1. The Workload Comparison

Proof of Lemma 1 of the manuscript. Put $D = U_{Q_1} - U_{Q_2}$. Arrivals add the same jump to both terms, so D is continuous, and between arrivals it changes at rate $-(\min(N_{Q_1}, k) - \min(N_{Q_2}, k))$; it is therefore piecewise linear with breakpoints among the finitely many arrival and completion instants of the two runs.

If $U_{Q_1}(u) > (k-1)L$ then more than $k-1$ jobs are present under Q_1 , each carrying at most L , so every server of Q_1 is busy and $\min(N_{Q_1}, k) = k \geq \min(N_{Q_2}, k)$: D does not increase at u . Fix t and let $S = \{u \leq t : U_{Q_1}(u) \leq (k-1)L\}$, which is non-empty because U_{Q_1} vanishes before the first arrival, and put $t_0 = \sup S$. The supremum need not be attained, since U_{Q_1} jumps upwards at arrivals, so we use the left limit: there are $u_n \uparrow t_0$ with $U_{Q_1}(u_n) \leq (k-1)L$, hence $U_{Q_1}(t_0^-) \leq (k-1)L$ and, D being continuous, $D(t_0^-) \leq U_{Q_1}(t_0^-) \leq (k-1)L$ because $U_{Q_2} \geq 0$. On $(t_0, t]$ we have $U_{Q_1} > (k-1)L$, so D does not increase there, and $D(t) \leq (k-1)L$. Exchanging Q_1 and Q_2 gives the other direction. That the constant cannot be lowered is the cascade of Section S8.1 of this file, which reaches $(k-1)L(1 - ((k-1)/k)^m)$ after m rounds. At $k = 1$ the bound reads $|D| \leq 0$ and is attained everywhere: one server executes work at rate 1 whenever any job is present, so $U_{Q_1} \equiv U_{Q_2}$. $\square$

Proof of the strict inequality in Lemma 1, for $k \geq 2$. Keep the notation above, so $D = U_{Q_1} - U_{Q_2}$ is continuous and piecewise linear with finitely many pieces. Suppose $D(t) = (k-1)L$ for some t and let t_1 be the smallest such t ; it exists because $\{D = (k-1)L\}$ is closed and $D(0) = 0 < (k-1)L$, which uses $L > 0$ and $k \geq 2$.

Let I be the linear piece immediately to the left of t_1 , which has positive length. Its slope is strictly positive: a negative slope would put $D > (k-1)L$ somewhere on I and a zero slope would put $D = (k-1)L$ on all of I , and either contradicts the minimality of t_1 .

The slope of D is $-(\min(N_{Q_1}, k) - \min(N_{Q_2}, k))$, so $\min(N_{Q_1}, k) < \min(N_{Q_2}, k)$ on I and therefore $N_{Q_1} \leq k - 1$ there.

With at most $k - 1$ jobs present and Q_1 work-conserving, none of Q_1 's jobs waits, so all of them are in service; hence U_{Q_1} falls at rate N_{Q_1} on I , and $U_{Q_1} \leq (k - 1)L$ on I because each of the at most $k - 1$ present jobs carries at most L . Taking left limits at t_1 and using the continuity of D ,

$$U_{Q_1}(t_1^-) - U_{Q_2}(t_1^-) = (k - 1)L, \quad U_{Q_1}(t_1^-) \leq (k - 1)L, \quad U_{Q_2} \geq 0,$$

which forces $U_{Q_2}(t_1^-) = 0$ and $U_{Q_1}(t_1^-) = (k - 1)L$. Since $L > 0$ and $k \geq 2$, this gives $N_{Q_1} \geq 1$ on I — with $N_{Q_1} = 0$ we would have $U_{Q_1}(t_1^-) = 0 = (k - 1)L$ — so U_{Q_1} is strictly decreasing on I and $U_{Q_1}(t_1^-) < \sup_I U_{Q_1} \leq (k - 1)L$, a contradiction. Exchanging Q_1 and Q_2 gives the other direction. Both steps that invoke $(k - 1)L > 0$ fail at $k = 1$, where the statement is false: there $U_{Q_1} \equiv U_{Q_2}$ and the bound 0 is attained at every instant. $\square$

S4.2. Consistency, and Why the Converse Fails

Proof of Proposition 1 of the manuscript. Induct on dispatch epochs: up to the first epoch at which the two runs could differ their completion histories agree, hence so does $\text{over}[\cdot]$, so $E(t)$ is empty in the wrapped run and it copies A . The first consequence uses $\text{over}[q](t) \leq \text{In}_q^A$, a completed overtaker of q having been dispatched while q waited; the second uses Corollary 2 of the manuscript. $\square$

The converse fails, so this is not an equivalence. Take $k = 1$, two jobs of service 1 at $t = 0$, the base policy “higher rank first”, and $B = 1$. Both runs dispatch in the order $(1, 0)$, yet at the second dispatch the waiting job has $\text{over} = 1 \geq B$, so the guard fires and happens to choose what the base would have chosen. Nor is there multiplicative consistency: with $k = 1$, one job of service L followed by m jobs of service 1 all at $t = 0$, the wrapper with $B = 0$ is first-come first-served while the base ordering by service time is not, and the ratio of mean waits is $(2L + m - 1)/(m + 1)$, unbounded in L .

S4.3. The Net-Overtake Identity, and Its Two Directions

This subsection proves Theorem 1 of the manuscript, of which Section 6 there keeps the statement, the busy-period accounting (1) and the idea; it then supports the statement made in the same section that the upward and downward extremes of the theorem do not have the same content, and that the executed-work convention would change only the downward one.

Proof of Theorem 1 of the manuscript. *Busy-period accounting.* Job i waits throughout $[a_i, s_i^P)$, so by work conservation every server is busy on that interval and exactly $k W_P[i]$ units of work are executed there, none of it on i . (If $s_i^P = a_i$ the interval is empty and both sides vanish.) Split that work by rank.

A job j of rank above i has $a_j \geq a_i$, so it executes inside the window only if it is dispatched inside it, that is only if $j \prec i$; conversely every such j is dispatched at or before s_i^P . The work executed on jobs of rank above i is therefore $\text{In}_i - \rho^{\text{new}}$, where ρ^{new} is the remaining work at s_i^P of those overtakers not finished by then; a same-phase overtaker contributes C_j to both terms and 0 to the executed work, as it must.

Every job of rank below i is present at a_i , with total remaining R_i^P . At s_i^P their total remaining is Out_i , the jobs not dispatched before i each carrying their full service time, plus ρ^{old} , the unfinished part of those dispatched strictly before i , so the work executed on them is $R_i^P - \text{Out}_i - \rho^{\text{old}}$. Adding the two parts gives equation (1) of the manuscript, $k W_P[i] = R_i^P + \text{In}_i - \text{Out}_i - \rho_i^P$ with $\rho_i^P := \rho^{\text{old}} + \rho^{\text{new}}$.

Bounding ρ . The jobs counted in ρ_i^P occupy distinct servers, and none of them occupies the server that takes i , so there are at most $k - 1$ of them and each has remaining work at most L : $0 \leq \rho_i^P \leq (k - 1)L$.

The reference policy. Under first-come first-served, $\text{In}_i = 0$, since a job of rank above i dispatched before i would have to be chosen at an instant at which i waits, and $\text{Out}_i = 0$, since at s_i^{FCFS} every job of rank below i has already been dispatched. So (1) reads $k W_{\text{FCFS}}[i] = R_i^{\text{FCFS}} - \rho_i^{\text{FCFS}}$ with $0 \leq \rho_i^{\text{FCFS}} \leq (k - 1)L$.

Comparison. At a_i , after the arrivals there are processed, the two systems contain the same jobs, and those of rank i or above are untouched in both, so $U_P(a_i) - U_{\text{FCFS}}(a_i) = R_i^P - R_i^{\text{FCFS}}$ and Lemma 1 of the manuscript gives $|R_i^P - R_i^{\text{FCFS}}| \leq (k - 1)L$, strictly for $k \geq 2$. Subtracting the two instances of (1) gives the identity. Each direction uses only two of the three ranges: upward $R_i^P - R_i^{\text{FCFS}} \leq (k - 1)L$ and $\rho_i^{\text{FCFS}} \leq (k - 1)L$ with $\rho_i^P \geq 0$, and downward the mirror image, so the slack is $2(k - 1)L$ and not $3(k - 1)L$. At $k = 1$ all three terms vanish. $\square$

Remark 1 (The upward and downward extremes have different content). *Upward, the extremal value is a statement about delay: on the witnesses we have computed exactly, none of In_i is same-phase work and the victim's own excess is about $+3L$. Downward, half of the extremal value is an accounting convention. There $\text{In}_i = (k - 1)L$ consists entirely of overtakers dispatched in the victim's own phase, which execute nothing while it waits; the same-phase share of $|\Delta_i|$ is 50% to 63% at $k = 2, 3, 4$ (Section S8.2), and the downward victim's excess is negative, since it starts earlier under P than under first-come first-served. If In_i is redefined as the work of higher-ranked jobs actually executed during $[a_i, s_i^P)$, Theorem 1 of the manuscript holds with the same constant and the same proof, the upward supremum is still $2(k - 1)L$, and the best downward witness we have falls to $(k - 1)L$. We keep the dispatch-sequence convention for two reasons. Its constant is exactly determined in both directions, whereas the downward constant for the executed-work variant is open, somewhere in $((k - 1)L, 2(k - 1)L]$; and its two sides are symmetric. Completed work lower-bounds both currencies, so the guard is implementable either way: a job j of rank above i that completes while i waits has $a_j \geq a_i$ and is dispatched at or after a_i , so all of C_j is executed inside $[a_i, s_i^P)$ and is counted in full by the executed-work variant as well.*

S4.4. What a Count of Overtakes Costs

This subsection supports Remark 1 of the manuscript, that a count bound on overtakes is sufficient for a per-job guarantee only through the worst-case charge of L per pass and is not necessary for one.

How the count is charged decides which guarantee. Charged when an overtaker is dispatched, every overtaker of i goes out at an instant at which no job fires, since a firing dispatch serves the queue head and the head is not an overtaker of i ; so at most κ overtakers are dispatched while i waits, which is what gives $\text{In}_i \leq \kappa L$ and equation (2) of the manuscript. Charging the count at completion instead leaves up to $k - 1$ overtakers in service and the one being dispatched uncounted, which costs a further L on the right and gives the baseline fewer passes at the same promise.

Both halves are below. The sufficiency uses the bound on service times, and without that bound a single overtake makes the excess arbitrarily large: at $t = 0$ let $k - 1$ jobs of service X , then a job of service ϵ , then one more job of service X arrive together; first-come first-served starts the victim immediately, a policy that takes the last job instead fills every server, and the victim waits X with an overtake count of 1. That instance steps outside Section 3 of the manuscript, which fixes $C_i \leq L$; inside the model the same accounting is what makes a count bound work, with L charged per pass. Necessity fails with no such caveat: normalise to $k = 1$, $L = 1$ and $G = 1/2$. Let a victim of service 1 arrive first together with M short jobs, each of service $1/(2M)$. Running the short jobs first gives

every job excess at most $1/2$, uses $\text{In}_i = kG$ exactly, and lets the overtake count M exceed any fixed limit.

S4.5. Proofs of the Guard Bounds

This subsection proves Theorem 3 and Theorem 4 of the manuscript, the additive guarantee of the wrapper under any budget rule that respects the cap and the multiplicative form the age-relative shape carries.

Proof of Theorem 3 of the manuscript. For (3) of the manuscript, let j be the overtaker of i dispatched last in the dispatch sequence and let t be its instant. Because dispatches are sequential and i is dispatched after j , job i is waiting at the moment j is dispatched; this includes the case $t = s_i^P$, in which j is dispatched in i 's own phase, just before i . Since j is dispatched, either $E(t)$ is empty or its minimum-rank member is j ; in both cases $i \notin E(t)$, because i has the smaller rank. Hence $\text{over}[i](t) < \text{budget}(i, t) \leq B_{\max}$.

The overtakers of i dispatched strictly before j split into those completed by t , whose total service is $\text{over}[i](t)$, and those still in service at t ; the latter occupy distinct servers, none of them the server taking j , so there are at most $k - 1$ of them, each of service at most L . Adding j itself gives $\text{In}_i < \text{budget}(i, t) + kL \leq B_{\max} + kL$.

For (4) of the manuscript, Theorem 1 of the manuscript with $\text{Out}_i \geq 0$ gives $k(W - W_{\text{FCFS}}[i]) \leq \text{In}_i + 2(k - 1)L$; substituting (3) of the manuscript and dividing by k gives the claim, strictly. If i has no overtaker then $\text{In}_i = 0$, the excess is at most $(2 - 2/k)L$, and the bound holds strictly because $L > 0$ in the model of Section 3 of the manuscript. $\square$

Proof of Theorem 4 of the manuscript. The argument for (3) of the manuscript gives $\text{In}_i < \text{budget}(i, t) + kL$ at the instant t of i 's last overtaker, and $t \leq s_i^P = a_i + W$ gives $\text{budget}(i, t) \leq B_0 + \eta kW$, which is (5) of the manuscript. Theorem 1 of the manuscript with $\text{Out}_i \geq 0$ then yields $kW - \eta kW < kW_{\text{FCFS}}[i] + B_0 + (3k - 2)L$; divide by k . Both inequalities are strict, by the same argument as Theorem 3 of the manuscript: when i has an overtaker the strictness comes from (3) of the manuscript, and when it has none $\text{In}_i = 0$ and the excess is at most $(2 - 2/k)L < (3 - 2/k)L$. $\square$

S4.6. Why the Multiplicative Bound Is Claimed for One Shape Only

This subsection supports the restriction Section 5 of the manuscript places on Theorem 4: with the queue-length shape the multiplicative bound stated with one common constant is false.

Remark 2 (Why the multiplicative bound is claimed for one shape only). *The same computation goes through when B_0 is replaced by a quantity $B_{0,i}$ fixed at i 's own arrival, and then bounds W in terms of $B_{0,i}$. The queue-length shape of Section 5 of the manuscript sets $B_{0,q} = B_0 + \gamma n_q$ with n_q the number of jobs waiting when q arrives, so $B_{0,i}$ varies from job to job. Stated with the common B_0 the multiplicative bound is then false, by a margin that grows with γ . Take $k = 1$, $B_0 = 0$, $B_{\max} = \gamma$, any $\eta \in [0, 1)$, and the input: a job of service L at $t = 0$, which is dispatched at once; a second job of service L at $t = 1$, which waits with $n_q = 0$; the victim i , also of service L , at $t = 1$ and ranked after it, so that i arrives behind a single waiting job, has $n_i = 1$ and hence budget γ ; and γ jobs of service 1 at $t = 2L$, all ranked above i . First-come first-served runs the first two jobs on $[0, L]$ and $[L, 2L]$ and starts i at $2L$, so $W_{\text{FCFS}}[i] = 2L - 1$. Under the wrapper the second job is taken at $t = L$ as well, by the guard if $\eta = 0$ and by the base otherwise; at $t = 2L$ the victim has $\text{over}[i] = 0 < \gamma$, so the guard does not fire and the base spends the whole budget on the stream, giving $W_P[i] = 2L - 1 + \gamma$ against a right-hand side of $(3L - 1)/(1 - \eta)$. The bound is violated for every $\gamma > (3L - 1)/(1 - \eta) - (2L - 1)$, that is for $\gamma > L$ at $\eta = 0$, and the margin grows without bound in γ . The same happens for any per-job constant. For $\gamma > 0$ we therefore*

claim Theorem 3 of the manuscript and not Theorem 4 of the manuscript. The cap, and with it the promise, is unaffected either way.

S4.7. Proofs of the Necessity Pair

This subsection proves Lemma 2 and Corollary 2 of the manuscript, which together make a work budget necessary for a per-job guarantee up to an additive $O(kL)$.

Proof of Lemma 2 of the manuscript. Every job j counted in Out_i has rank below i , so first-come first-served dispatches it before i and $s_j^P \leq s_j^{\text{FCFS}} + G \leq s_i^{\text{FCFS}} + G$; it is also dispatched at or after s_i^P . All of them therefore start inside a window of length $w = G - \text{excess}_P[i] \geq 0$. Of the jobs one server starts inside such a window, all but the last complete inside it, so their total service is at most w and the last adds at most L . $\square$

Proof of Corollary 2 of the manuscript. Theorem 1 of the manuscript gives $\text{In}_i - \text{Out}_i \leq k \text{excess}_P[i] + 2(k-1)L$; add Lemma 2 of the manuscript, whose hypothesis holds because the guarantee is assumed for every job, and the terms in $\text{excess}_P[i]$ cancel. $\square$

S4.8. The Floor of Order L , and the Scope of Its Hypothesis

This subsection proves Proposition 3 of the manuscript and shows that its hypothesis can be neither dropped nor tightened, the remaining cases of the budget family being exhaustive.

Proof of Proposition 3 of the manuscript. Into an empty system, at $t = 0$, let $k+1$ jobs arrive together, all of service L . The first k ranks are precisely the jobs FCFS starts at once. Nothing has completed, so every waiting job has $\text{over} = 0$; because its first-epoch budget is positive, no waiting job fires and the wrapper accepts the base policy's choices. At the first choice of a job beyond the first k ranks, that job displaces one of FCFS's first k equal-length jobs. The displaced job waits until a completion under the wrapper but starts at zero under FCFS, and hence has excess L . A prediction-ranked base produces the same choice when one of the later ranks receives the preferred score. At $k = 1$ and $B > 0$, (3) of the manuscript gives excess $< B + L$ strictly, and a family approaching it is in Section S8.4; $B = 0$ is FCFS. $\square$

Two remarks on the hypothesis, because the proposition is easy to state too widely and too narrowly. It cannot be dropped: as a claim about policies the statement is false, since first-come first-served is non-preemptive, work-conserving, reads no service time before completion, and has excess 0 on every input, as does any rank-ordered policy. For the same reason it is false as a claim about every base policy: the guard run around first-come first-served is first-come first-served, whatever the budget. The force comes from the guard's not firing, which is a property of the budget rule and not of work conservation, together with a base policy that uses the freedom this leaves it. It is also not necessary. A rule that is 0 at the first dispatch epoch and positive later, such as the age-relative shape with $B_0 = 0$ and $\eta > 0$, falls to the same instance placed behind a saturating prefix: let k jobs of service L arrive at $t = 0$, the victim just after them, and k further jobs just before $t = L$. The victim's wait up to $t = L$ is what first-come first-served gives it, its budget at $t = L$ is positive because it has aged, and the k newcomers then take every server. The remaining case is degenerate: if the rule gives budget 0 to a job whenever no completed work has overtaken it, then at every dispatch epoch the lowest-ranked waiting job h has $\text{over}[h] = 0$, so h lies in $E(t)$ and is its minimum-rank member and is dispatched, and by induction the wrapper is first-come first-served itself, whatever the base policy. For the budget of Algorithm 1 of the manuscript these cases are exhaustive. If $B_{\max} = 0$

or $B_0 = \gamma = \eta = 0$ the budget is identically zero and the wrapper is first-come first-served. If $B_0 > 0$ the proposition applies, and if $B_0 = 0 < \eta$ the saturating prefix does. If $B_0 = \eta = 0 < \gamma$, let $k + 2$ jobs of service L arrive together at $t = 0$: the first has $n_q = 0$, hence budget 0, and is released at once, every other job has a positive budget and nothing has overtaken it, so a base policy that prefers the highest ranks passes over the second job at $t = 0$ when $k \geq 2$, and at $t = L$ when $k = 1$, and that job's excess is L . A budget rule of this family therefore either makes the wrapper first-come first-served around every base policy, or admits a base policy and an input with excess at least L .

S4.9. Why the Budget Is Measured in Work and Not in Positions

This subsection supports the choice of currency Section 5.2 of the manuscript makes, and the looseness Section 8.5 measures.

Earlier bounded-overtaking mechanisms limit how many times a job may be passed, which is the currency of the finite-skip and Nudge- K rules [8–10]. Counting is not necessary: a job can be passed any number of times and stay inside its promise, provided the work that passes it is small. Counting is sufficient in this model, but only through the worst-case charge of L per overtake, since every overtaker is bounded by the time limit; a count of κ therefore buys a promise of $(\kappa + 2k - 2)L/k$ (Remark 1 and equation (2) of the manuscript). That is the bound a count can give, and it is loose by the ratio $L/\bar{C}$, one to two orders of magnitude on our traces. Ten overtakers that each run for the full limit L and ten that run for ten milliseconds carry the same count and delays that differ by three orders of magnitude, so the charge is paid on every dispatch whether or not the work is there. Section 8 of the manuscript runs the count rule as a baseline and measures how much the looseness costs on a real load. That work is the currency the excess is actually written in is the content of the identity of Section 6 of the manuscript, not a claim of ours about what earlier mechanisms could not do.

S5. Reproducibility and Provenance of the Empirical Results

S5.1. Feature Definitions, Targets and Rolling Splits

The executable feature dictionary is `src/spjf_guard/features/sweep.py`: `HIST_COLS`, `CTX_COLS`, `GROUPS` and `feature_frame` specify the entity aggregates, missing values and event order; `src/spjf_guard/data/events.py` specifies code columns and clock inputs. M4 combines static code, assessment/calendar context, exercise history, user history, user-exercise history and previous-event history. It excludes the relational extensions. Outcome aggregates contain log-cost moments and quantiles, error/heavy rates and recent-result summaries; counts and elapsed time since arrival are known at arrival and are distinguished from outcome fields. The rolling-history window, tie order and update formulas are defined in that dictionary. Static uses only the code and context blocks. The independent source-clock audit in Section S1 verifies that dictionary under its original clock, not under an arbitrary replayed policy.

Package scores are fitted by `src/spjf_guard/predict/forward.py`. For each target term, training includes only terms whose first arrival precedes that target, and only records whose result is readable strictly before its first arrival. History thresholds and user-slowness cutpoints are refitted from those training rows; a model is then frozen for the target term. The targets are 2020-ERE, 2020-2, 2021-1, 2021-2, 2022-1 and 2022-2. Starting at 2018-1, their training periods end respectively at 2019-2, 2020-1, 2020-2, 2021-1, 2021-2 and 2022-1; availability filtering still applies at each cutoff. Validation overlays exclude the final development term from selection, but that term and the earlier family exploration are development data, not sealed testing.

The score objectives are in `src/spjf_guard/predict/scores.py`, and the deterministic fit settings in `lgb_parameters` and the configuration identified by the output manifest. `spjf_e` is a Tweedie regression of capped seconds; `spjf_log` is squared-error regression of log-transformed cost. In contrast, the earlier M4 AUROC 0.9283 [0.9118, 0.9415] is the heavy-class classifier probability, averaged over 3 seeds, trained on 2018-1–2019-2 and tested only on 2022-2. Those retrospective classifier predictions never enter scheduling. The package's pooled `spjf_e` AUROC 0.9360 combines 402,358 unreplicated forward-prediction rows. Its evaluation heavy threshold stays at the core training p95, 1.559043 s; it is not the threshold refitted inside each model's history features. Predictor intervals resample users, whereas scheduling intervals resample weeks. The two AUROCs compare different fits, targets and pooling and must not be read as a measured improvement on a common split.

S5.2. Exact Selection Grids and Counts

The full expanded grid, its definitions, counts and deduplication totals are in `outputs/selection_v3/selection_protocol.json`; per-cell values, worst-cell values and selections are in the adjacent `selection_grid.csv`, `selection_worst.csv`, `selection_frontier.csv` and `selected_parameters.csv`. All budget and queue-work bases below scale by $k/4$ before dispatch arithmetic. The cap is determined by the promise.

The constant grid has the following 39 base budgets (seconds), shared across promises:

15, 17.99, 21.576, 25.877, 30, 31.035, 37.222, 44.641, 53.54, 60, 64.212, 77.012, 92.363, 110.774, 120, 132.855, 159.338, 191.099, 229.192, 240, 274.878, 329.671, 360, 395.386, 474.201, 480, 568.726, 600, 682.093, 818.058, 981.126, 1176.7, 1411.26, 1692.57, 2029.96, 2434.6, 2919.9, 3501.94, 4200.

It is the union of 32 logarithmically spaced values between 15 and 4200 s, rounded as recorded, and the anchors 15, 30, 60, 120, 240, 360, 480, 600, with duplicate values removed. A full-cap constant point is added at each of $G = 300, 600, 1200$ s. This makes 42 expanded constant entries.

For each promise the age-relative grid is the Cartesian product

$$B_{0,\text{base}} \in \{0, 15, 30, 60, 120, 240, 360, 480, 600\}, \quad \eta \in \{0, 0.25, 0.5, 0.75, 0.9, 0.95\}, \quad \gamma = 0. \quad (750)$$

The queue-length grid is

$$B_{0,\text{base}} \in \{0, 30, 120\}, \quad \eta \in \{0, 0.9\}, \quad \gamma_{\text{base}} \in \{1, 4, 16\}. \quad (752)$$

Thus the family counts over all promises are $42 + 162 + 54 = 258$ expanded points. Deduplicating identical dispatch rules after scaling and saturation leaves 243 distinct schedules at each evaluated server count. This is a computational deduplication, not a reduction of the declared search space.

Candidate counts per promise answer a different question. The constant entries shared across promises qualify for a given G only if their worst-cell certified promise is at most G ; an age-relative or queue-length entry belongs to its own declared promise. Feasibility then additionally requires $\text{harm} \leq G/2$ in every validation cell. Table S15 separates these filters, explaining why the per-promise totals differ from the expanded search size. The objective is the minimum deadline gap over all validation cells, with ties resolved by smaller worst harm, then smaller B_0 , η and γ .

The aging comparator has the credit grid 0.00001, 0.00003, 0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3 and 1. It uses the same validation pool, harm threshold and worst-cell objective, with ties to less harm then larger credit. It uses conservative scores; the guard

Table S15. Candidates / feasible points after promise assignment and the harm filter.

G (s)	Constant	Age-relative	Queue-length	Joint
300	27 / 22	54 / 15	18 / 7	99 / 44
600	35 / 31	54 / 30	18 / 9	107 / 70
1200	39 / 35	54 / 43	18 / 17	111 / 95

grids used the original package score. The selected guard parameters are subsequently held fixed across visibility controls, so this is not a claim of reselection under conservative or exact histories. The aging label carries no proved guarantee.

S5.3. Replication, History and Aggregation Order

Each overlay is built from the same 40 class-terms replicated 44 times with independent whole-week shifts; each of the five overlays therefore reuses the same jobs. Original scores are computed on the source calendar and copied with their source jobs. Conservative histories are namespaced within a class-term, use only replayed submissions and share the copy’s shift. These copies are not independently trained predictors or independent workload samples.

For the pending exact refinement, a copy means the same class-term, replication round and shift. Its own submission outcomes follow policy-specific completions. Other classes and terms are exogenous history in that copy’s original relative clock: when the target copy shifts by s_i , an external result is interpreted at $\text{done}_j + s_i$, not at another copy’s independent shift. This preserves earlier completed history without claiming a single platform whose copied users share dynamically changing state. A separate sensitivity removes entire records from other replayed classes in the same term, including arrival counts and elapsed time since arrival; it then repeats same-copy refinement. Ordinary outcome withholding preserves already-observed arrivals. The event order releases positive duration completions before reads at the same instant, and zero-duration outcomes after those reads. These conventions are specified in `docs/policy_consistent_visibility.md` and implemented by `src/spjf_guard/features/refine.py`.

In the package main and visibility tables, each overlay’s deadline p99 is first computed over its eligible jobs; its gap is then computed against that overlay’s FCFS and SJF. Reported p99 and gap are separate arithmetic means over overlays. Consequently the displayed mean waits need not algebraically recover the displayed mean gap. Harm and maximum excess are maxima over overlays. Firing is queue-weighted within a cell, then averaged across overlays. Each bootstrap draw uses the same week multiplicities for every policy and overlay, recomputes weighted job quantiles and cell-level ratios, and averages the ratios. No table uses an average of weekly p99 values. The capacity-sweep gap instead uses the ratio of averaged overlay quantiles, as explicitly identified in Section S6.

S5.4. Which Results the Package Reproduces

The package produces the main CodeBench ranking, guard, adversarial and residual tables from `outputs/dev_tables/`, including its separate single-server directory; predictor metrics from `outputs/dev_predictor/`; and visibility comparisons from `outputs/dev_visibility/`. Guard selection and aging selection are in `outputs/selection_v3/` and `outputs/selection_aging/`. Their manifests identify inputs, settings and output hashes. The preserved original recipe is `configs/main_original_84932d9.yaml`; the original outputs are not claimed to have been regenerated under an unfinished visibility revision.

Earlier predictor-family comparisons and source-clock feature sensitivities are archived under `evidence/codebench_service_v2/` and `evidence/predictor_neural/`. The refit-control and GRU rows in the main ranking table come from `evidence/main_v3/`;

the ranking-loss table uses its own fits in `evidence/ranking_score/`. Cross-domain rows combine the package CodeBench row with `evidence/cross_domain/`, `evidence/accoding_v2/`, `evidence/firefox_ci/` and the LPC-EGEE study below. CI chronological-refit validation is in `evidence/firefox_ci_holdout/`. The capacity and physical experiments are separate analyses in `prechecks/weakness1_attack/`.

The trace-construction sensitivity table (Table 12 in the unrevised numbering) uses the earlier source-clock M4 log-scale score, overlays 0–2, and a fixed completed-work budget $B = 600$ s. This is neither a 600-second promise nor the selected age-relative Guard(600). It supports the earlier ordering result's sensitivity to term inclusion, the alternate timestamp interpretation, and merging or dropping co-ending runs. It does not test the final expected-cost guard, conservative history or policy-specific refinement under those altered inputs. Numerical agreement of the exact-arithmetic package and its independent reference is also distinct from the archived earlier-pipeline validation.

The sealed procedure is specified by `docs/sealed_run_procedure.md`. It produces separate multi-server, single-server, predictor and visibility outputs, including the policy-specific exact stage, with unchanged declared selection rules and parameters. It does not extend the earlier predictor-family or cross-domain studies. No sealed result is present in this supplement.

S5.5. The Selection Rule and the Cost per Event

This subsection supports Section 5.2 of the manuscript: it states the pre-stated selection rule in full and gives the data structures that make the guard's firing test cost $O(\log n)$ per event.

The rule is fixed before the grids are run: three pre-stated grids, one per shape, 258 points in all; a point is feasible when its harm is at most $G/2$ in every one of the 15 validation cells; the objective is the deadline-window gap closed in the worst cell, with ties going to smaller worst harm and then to smaller B_0, η, γ . Guard(G) is the winner over the three grids together; the winner inside each grid is reported as an ablation row.

Algorithm 1 of the manuscript is written for readability rather than for execution, since its inner loop over waiting jobs at each completion is linear. We keep $\text{over}[q] = TC - F(q)$ instead, where TC is the total completed work and $F(q)$ the prefix sum over ranks up to q of completed work, maintained in a Fenwick tree, so that a completion is one point update and a query one prefix sum. The firing test is a segment tree over the window of waiting ranks supporting suffix range-add and the query "leftmost leaf with value at least T ", after rewriting $\text{over}[q](t) \geq B_{0,q} + \eta k(t - a_q)$, where $B_{0,q} = B_0 + \gamma n_q$ is fixed at q 's arrival, with $\eta k = e_n/e_d$ in lowest terms as $e_n a_q - e_d B_{0,q} - e_d F(q) \geq e_n t - e_d TC$, whose left side is a per-job value changed only by range-add at completions and whose right side depends on t alone. The capped channel needs only the head, since at a fixed threshold over is non-increasing in rank. All budget arithmetic is carried in integer microseconds, so the test is exact and no floating-point clock enters the schedule. Each event costs $O(\log n)$ in the guard's own maintenance, exclusive of what the base policy spends producing a proposal, and n counts waiting jobs, with arrivals sharing a timestamp processed in rank order.

S5.6. What the Selection Was Run On

This subsection supports the three statements Section 8.2 of the manuscript places before its numbers, by recording what the released package regenerates and what remains archived development evidence.

Third, the grids, the feasibility rule, the objective and the tie-break were fixed before the grids were run, at comparable density across the three shapes, and the package com-

parisons use exact integer microseconds for arrivals, service and budget arithmetic. Earlier predictor comparisons, cross-domain analyses and trace-construction sensitivities remain archived development evidence, with their different fits and configurations identified in the captions and this section. They are not all regenerated by the package. Two properties of the outcome belong here. The winning shape is not the same at every promise: the age-relative shape wins the joint comparison at $G = 300$ and 600 s and the queue-length shape at $G = 1200$ s, where it reaches worst-cell gap 0.7283 at worst-cell harm 145.3 s against 483.9 s for the best age-relative point. And a worst-cell rule can be decided by a single cell: dropping the one validation cell that decides the $G = 600$ choice moves that choice to $B_0 = 15k/4$ s, $\eta = 0.90$, with worst-cell gap 0.7237. The sealed terms will be opened once after the protocol freeze and reported in the same multi-server tables as an additional column, with the parameters unchanged. The current frozen-run procedure also produces a separate single-server trace and its tables, per-term and pooled predictor metrics, conservative/static visibility tables and exposure/lag audits, and a separate policy-specific exact-visibility stage. It does not produce the refit-control and GRU rows of Table 4 of the manuscript, the other predictor-family comparisons, cross-domain sealed results, or the earlier trace-construction sensitivities. No sealed result is printed here.

S6. Capacity, Physical Queueing and Feedback

These development experiments address distinct questions. The capacity sweep changes the size of the constructed pool while preserving its complete offered stream. The physical experiment checks an executing timed-work service on a selected window. Neither identifies a historical queue on CodeBench or a causal deployment effect. The scores are the archived embedded Tweedie predictions, not the later package refit, conservative score or pending exact visibility variant.

S6.1. Capacity Sensitivity on Complete Traces

The declared common grid uses every integer $k = 1, \dots, 20$ on all five development overlays; overlay 0 alone continues through $k = 76$. This gives 156 capacity cells. Each replay starts at the beginning of its complete trace, keeps queue state across weeks and drains all jobs; there is no time compression, window reset or thinning. The same source jobs occur in all overlays, so these are alternative constructions rather than independent deployments. The grid and comparison family were declared after a pilot on the first overlay, before the full sweep, not before inspecting the development demand.

Table S16 reports the average of the overlay-specific deadline p99 values, with point-wise intervals. Gap closure in this study is the ratio of differences of those averaged quantiles. This differs from the package main tables, which average overlay-specific gap ratios. Paired calendar-week resampling uses 2,000 common draws, recomputing quantiles from the weighted job distribution, not averaging weekly quantiles. The declared simultaneous band covers 80 absolute FCFS-minus-policy differences: each capacity and each non-oracle policy. It uses a centred coordinate-standardised maximum statistic. These intervals resample realised outcomes, not queue trajectories; they omit source-term selection, predictor fitting, shared source jobs, model mismatch, feedback and cross-week state dependence. Maxima below are trace maxima, not population confidence bounds.

The negative cells are material. At $k = 1$, FCFS-minus-Guard p99 differences are -208.61, -493.79 and -1,071.72 s for the 300, 600 and 1200 s promises; all simultaneous bands are wholly negative. At $k = 2$ all guard point estimates remain negative. Positive simultaneous lower bounds occur for SPJF-E at $k = 2$ –12, Guard(300) and Guard(600) at $k = 4$ –12, and Guard(1200) at $k = 3$ –12. At $k = 14$ the guard point improvements are about 0.37 s

Table S16. Capacity sweep with archived scores: mean overlay deadline p99 in seconds, with point-wise paired calendar-week intervals. Negative effects are retained.

<i>k</i>	FCFS	SPJF-E	Guard(600)
1	108,345.31 [92,267.38, 114,125.36]	130,231.12 [65,565.87, 155,632.22]	108,839.10 [92,739.31, 114,582.34]
2	10,362.62 [7,565.14, 11,503.45]	1,844.21 [766.70, 3,990.37]	10,618.06 [7,820.32, 11,755.23]
3	1,056.40 [688.91, 1,273.01]	138.86 [105.79, 169.05]	934.68 [452.24, 1,253.73]
4	273.50 [197.09, 352.10]	62.92 [53.69, 72.13]	89.61 [65.94, 114.14]
5	119.35 [92.91, 143.43]	43.39 [37.94, 47.54]	47.44 [40.46, 53.39]
8	24.43 [18.46, 29.58]	13.21 [9.58, 15.80]	13.36 [9.67, 16.05]
14	0.84 [0.45, 1.50]	0.47 [0.30, 0.73]	0.47 [0.30, 0.73]
20	0.00 [0.00, 0.04]	0.00 [0.00, 0.03]	0.00 [0.00, 0.03]

but their simultaneous bands cross zero; at $k = 20$ the point differences are zero. A correct per-job bound does not imply positive tail benefit across queueing capacities.

For Guard(600) at $k = 4$ the maximum excess is 515.84 s and harm is 339.06 s, although it closes 0.8000 [0.7620, 0.8388] of the reference gap. At $k = 20$ that ratio is undefined because the reference gap is not positive, while maximum excess and harm remain 124.38 and 37.37 s. On overlay 0, FCFS still has one positive wait at $k = 75$; at $k = 76$ every policy has zero wait. The latter is the exact maximum infinite-server occupancy, hence a zero-wait endpoint for this overlay, not a capacity recommendation for a deployment.

S6.2. An Executing Timed-Work Service

The physical selection rule chooses the complete calendar-aligned 300-second bin with the greatest offered capped work in development overlay 0, breaking ties by earliest time. Its 2,151 jobs carry 4,118.215007 s of requested work, 6.864 times the two-worker release-window capacity. Each policy starts empty and drains its own queue. Inter-arrival spacing, requested costs and scores are retained; workers hold their slots with timed sleeps. This tests queueing and the control rule, not the portability of submitted-program execution time. It also discards inherited full-trace backlog, so its percentile is not the capacity sweep’s percentile.

The nominal payload cap is 60 s; the measured-service premise $L = 61$ s was fixed before execution. Guard parameters are $B_0 = 30$ s, $\eta = 0.5$, $\gamma = 0$ and $B_{\max} = 356$ s, corresponding to the nominal 300-second ideal-model promise. A measured holding time above L would fail the premise, not be clipped or used to raise L retrospectively. None did. Actual enqueue, choice, dispatch, worker start/finish and completion receipt are recorded separately; service becomes known only at completion receipt.

Table S17. Physical timed-work runs. The ideal comparator uses each run’s own measured arrivals and holding costs; all entries are seconds.

Policy	Mean wait	p99 wait	Max absolute per-job ideal discrepancy
FCFS	1,337.05	1,763.30	0.49
SPJF-E	142.79	1,687.47	392.84
Guard(300)	480.39	1,783.65	1,354.10

All 6,453 observed-prefix decisions passed an independent choice audit. Physical and ideal measured-input Guard firing counts were 322 and 372. The small mean discrepancies do not establish job-level fidelity: the maximum absolute per-job error is 392.84 s for SPJF-E and 1,354.10 s for Guard. Guard’s physical p99 exceeds the separate physical FCFS run by 20.34 s. Thus the physical window confirms an executing queue and rule conformance but does not show a guarded p99 gain. Actual arrivals and holding costs differ between

physical runs; their comparison is descriptive. One selected window supplies no across-window confidence interval.

The guard's theorem reference is shadow FCFS on that guard run's measured arrivals and service, not the separately timed physical FCFS run. An observed-trace certificate augments the ideal bound by cumulative idle worker-time and dispatch-to-start delay. Its largest correction is 1.127 s; all 2,151 jobs pass, and the largest uncorrected excess over shadow FCFS is 222.21 s. This is an ex post certificate conditional on the measured trace and the 61 s service premise. It does not bound future operating-system stalls and does not restore an advance uncorrected real-time promise.

S6.3. Why Open-Loop Benefit Has No Known Bias Direction

The replay compares policies conditional on fixed offered work. Feitelson states the objection at its strongest, that "logged workloads actually contain a 'signature' of the logged system" [11]: a trace-driven simulation issues requests irrespective of system state [12], so faster replies cannot induce more submissions, change their costs or alter which jobs enter a deadline-window quantile. Shmueli and Feitelson also distinguish increased productivity from improvement in conventional waiting metrics [12]. The two constructions below show that the bias from the missing feedback has no known direction.

In an exact single-server feedback construction, jobs of length 10 and 1 s arrive together, the long job first, and the short job's user submits a second unit job 0.5 s after completion. Replaying the FCFS-induced arrivals gives SJF a mean-wait advantage of 3.00 s; letting each policy generate its own successor arrival changes the advantage to -0.167 s. The corresponding linearly interpolated p99 advantages are 8.82 and 0.47 s. The job cohort is identical, so this example does not depend on comparing different populations.

The opposite bias direction is possible too. An initial FCFS order with services 20, 1, 3 and 10 s, whose unit-job user returns after 2 s with a 0.1 s successor, gives open- and closed-loop mean advantages of 9.20 and 11.16 s and p99 advantages of 10.00 and 10.18 s. These exact recursions falsify a general lower-bound interpretation; they are not fitted behavioural models. The pathwise theorem still compares a guarded schedule with shadow FCFS on that schedule's realised arrivals and services, not independently evolving closed-loop systems [12].

S6.4. LPC-EGEE: Recorded Queueing with Weak Counterfactual Validation

The public LPC-EGEE trace supplies recorded waits and concentrated execution costs together [13]. After outage exclusion and queue-limit capping, the longest 1% of 206,222 retained jobs carries 40.7% of work. On the chronological evaluation part the top share is 35.1%, which is the value used in the main cross-domain table. The physical pools stay disjoint. Effective capacities 83 and 56 are fitted only on the earlier part, with no test-driven refit. Service is the minimum of recorded run time and its documented queue limit; the common bound uses the largest pool limit, 259,200 s.

The fitted FCFS approximation underestimates evaluation-part mean waits by 24% and 39%, with hourly correlations of 0.44 and 0.40. The archive describes plain FCFS serial-job ordering during the logged period, but queue concurrency ceilings, polling and middleware are absent from the work-conserving replay. Recorded waits therefore cannot be equated with the theoretical FCFS counterfactual. This is weak simulator validation, not evidence that the original scheduler was an arbitrary non-FCFS policy.

In the conditional replay, SPJF-E reduces mean wait by 21.4% [13.2%, 38.2%], but its busy-hour p99 reduction of 4.2% [-11.2%, 46.0%] remains unresolved. Busy hours use frozen arrival-count thresholds, not deadlines. All six constant/age work-budget guards at the tested multiples of L fire zero times; a small fraction of the numerical bound being

spent is therefore not evidence of effective protection. Histories use recorded completion times, and 2,906 evaluation arrivals (2.5%) have an outcome unavailable under replayed SPJF-E. This row is a recorded-feature conditional counterfactual, not a closed-loop deployment result.

S7. Additional theory: scope, accounting, and instance-dependent bounds

This section records results that delimit, rather than enlarge, the main-text claims. The first example rules out instance optimality for the uniform guard. The exact identity that follows is a useful job-level accounting decomposition; its core is standard sample-path workload balance, and we do not treat it as a new conservation law. The final bounds replace the global timeout by realised order statistics, but in their actual-size form they are retrospective certificates under completion-only observation.

S7.1. Comparison with the nearest guarantee disciplines

Table S18. Scope of the nearest per-job guarantees and wrapper precedent.

Method	Reference discipline	discipline	Arrival assumptions	as-	Preemption	Information needed	Server model	Guarantee
This work	Same-input FCFS		Arbitrary finite path	fi-	No	Ranks, completions, opaque base proposal	k identical servers	Deterministic per-job additive waiting excess; wrapper around any base
PFCFS [14]	Flow time with-out later submissions		Online leases	re-	Gang pre-emption	Processor demand; service time need not be known; pre-emption penalty	Rigid jobs	λ -fairness; PFCFS is $(2 + 2s)$ -fair with competitive efficiency bounds
Conservative backfilling [15]	Submission-time reservation		Online missions	sub-	No	Requested time and resources	Rigid jobs	Later backfills do not delay any reserved start
PV-EASY [16, 17]	Priority reservation		Online missions; prediction error allowed	sub-	Kill and restart	Runtime estimates and resources	Rigid jobs	Zero lower-priority delay to the protected reservation
Nudge [8]	FCFS		Stationary Poisson		No	Size class at arrival	One server	Response-time stochastic dominance for its model; at most one swap per job
Load balancing Guardrails [18]	Bal-Unconstrained dispatcher		Stochastic heavy-traffic regime	SRPT	at servers	Size class and dispatch state	Parallel pools	Any dispatcher becomes mean-response-time optimal in heavy traffic

S7.2. No uniform instance-optimality ratio

Proposition 1 (An unbounded instance-optimality ratio). *At $k = 1$, let three batch jobs have rank-order service times $(L, 1, 1)$, with $L \geq 2$, and set $G = L$. The cap of the parameter rule of the main text (its Corollary on parameters from a promise, $B_{\max} = k(G - (3 - 2/k)L)$) is $B = 0$, so the guard is FCFS and has total waiting time $2L + 1$. The order $(1, 1, L)$ has total waiting time 3 and satisfies the per-job promise G . Consequently the ratio between the guard's total wait and that of a feasible policy is at least $(2L + 1)/3$, which is unbounded as $L \rightarrow \infty$.*

Proof. FCFS produces waits $0, L, L + 1$. Under the competing order the two short jobs wait 0 and 1, and the long job waits 2. The long job's excess is therefore 2 and both short jobs have nonpositive excess, so $L \geq 2$ makes the order G -feasible. The comparison allows a clairvoyant feasible policy; restricting the comparator to completion-only policies would be a different question. $\square$

The pointwise-maximal budget $B^* = G - L + 1$ of the main text's one-server theorem admits one short job here, giving total wait $L + 2$ and ratio $(L + 2)/3$; the conclusion is unchanged.

S7.3. Idle-capacity accounting

Let t_0 be the first arrival time, let $b_Q(u)$ be the number of busy servers under policy Q at time u , and define the cumulative unused capacity

$$\Gamma_Q(t) = \int_{t_0}^t (k - b_Q(u)) \, du.$$

This includes idle capacity caused by having fewer than k jobs in the system; work conservation excludes only deliberate idling while a job waits.

Lemma 1 (Exact accounting with idle-capacity correction). *For every finite input, every non-preemptive work-conserving policy P , and every job i ,*

$$k(W_P[i] - W_{\text{FCFS}}[i]) = \text{In}_i - \text{Out}_i + \Gamma_P(a_i) - \Gamma_{\text{FCFS}}(a_i) - \rho_i^P + \rho_i^{\text{FCFS}}.$$

The identity was checked on 214,735 job-instances, including tied events, non-batch arrivals, and same-phase overtakers.

Proof. Let $A(t)$ be cumulative arrived work and $E_Q(t)$ cumulative executed work. Then

$$U_Q(t) = A(t) - E_Q(t) = A(t) - k(t - t_0) + \Gamma_Q(t).$$

At a_i , after simultaneous arrivals and before that phase's dispatches, work from jobs not ranked below i is identical in the two runs. Hence

$$R_i^P - R_i^{\text{FCFS}} = U_P(a_i) - U_{\text{FCFS}}(a_i) = \Gamma_P(a_i) - \Gamma_{\text{FCFS}}(a_i).$$

Substitution into the two busy-period equalities of the main text (the displayed equality $k W_P[i] = R_i^P + \text{In}_i - \text{Out}_i - \rho_i^P$ in the proof of the identity), one for P and one for FCFS, proves the formula. A same-phase overtaker enters both In_i and ρ_i^P with its full size and therefore cancels. $\square$

Writing $D_i = \Gamma_P(a_i) - \Gamma_{\text{FCFS}}(a_i)$, the arrival-frozen budget

$$b_i = \max\{0, k(G - L) - (k - 1)L - D_i\}$$

has the mathematical promise G in the standard range $G \geq (3 - 2/k)L$: combine $\text{In}_i < b_i + kL$ with Lemma 1 and $\rho_i^{\text{FCFS}} \leq (k - 1)L$. This is not a deployable rule under the paper's completion-only observation model. In general $\Gamma_{\text{FCFS}}(a_i)$ depends on when an FCFS-shadow job completes before its size has been revealed in the realised run; simulating the shadow does not supply that missing completion time.

S7.4. Instance-dependent order-statistic bounds

For a set of jobs S , let $\Lambda_r(S)$ be the sum of its largest at most r original service times, with missing terms replaced by zero. We abbreviate

$$\Lambda_{k-1}(t) = \Lambda_{k-1}(\{j : a_j \leq t\}), \quad \Lambda_{k-1}^{<i} = \Lambda_{k-1}(\{j : j < i\}),$$

and

$$\Lambda_k^{>i}(t) = \Lambda_k(\{j : j > i, a_j \leq t\}).$$

Proposition 2 (Instance-dependent workload gap). *For any two non-preemptive work-conserving policies Q_1, Q_2 ,*

$$|U_{Q_1}(t) - U_{Q_2}(t)| \leq \Lambda_{k-1}(t) \quad \text{for every } t.$$

Proof. The workload difference is continuous across arrivals, whereas $\Lambda_{k-1}(t)$ can only increase. If $U_{Q_1}(t) > \Lambda_{k-1}(t)$, policy Q_1 must have at least k jobs present: otherwise their remaining work is no larger than the sum of their at most $k-1$ original sizes. All its servers are therefore busy, so the difference cannot increase. It cannot cross the nondecreasing barrier from below; exchange Q_1 and Q_2 for the other direction. $\square$

Proposition 3 (Instance-dependent excess accounting). *For every policy P and job i ,*

$$k \text{ excess}_P[i] \leq \text{In}_i - \text{Out}_i + \Lambda_{k-1}(a_i) + \Lambda_{k-1}^{<i}.$$

Proof. In Lemma 1, Proposition 2 bounds the idle-capacity difference, equivalently the workload difference at a_i , by $\Lambda_{k-1}(a_i)$. Drop $-\rho_i^P \leq 0$. The at most $k-1$ FCFS jobs still running when i starts all have rank below i , so $\rho_i^{\text{FCFS}} \leq \Lambda_{k-1}^{<i}$. $\square$

Proposition 4 (Instance-dependent overtaking under the guard). *If job i has an overtaker and τ is the dispatch time of its last overtaker, then*

$$\text{In}_i < \text{budget}(i, \tau) + \Lambda_k^{>i}(s_i^P).$$

If the budget is nondecreasing in time, the first term may be replaced by $\text{budget}(i, s_i^P)$; with only a uniform cap it may be replaced by $B_{\max}$.

Proof. Completed overtaking work before the last overtaker is dispatched is strictly below $\text{budget}(i, \tau)$. At τ , at most k dispatched overtakers remain unfinished: the newly dispatched job and at most one on each other server. Their total original size is at most $\Lambda_k^{>i}(s_i^P)$. $\square$

Proposition 5 (Combined instance-dependent guard bound). *Under a cap $B_{\max}$, every job satisfies*

$$k \text{ excess}_P[i] \leq B_{\max} + \Lambda_k^{>i}(s_i^P) + \Lambda_{k-1}(a_i) + \Lambda_{k-1}^{<i},$$

and the inequality is strict when i has an overtaker. When it has none, the sharper weak bound omits the first two terms.

Proof. Combine Propositions 3 and 4, use $\text{Out}_i \geq 0$, and apply the cap. If no job overtakes i , then $\text{In}_i = 0$ and Proposition 3 gives the stated sharper form. $\square$

These Λ quantities use actual sizes, including sizes of unfinished jobs, and hence are offline instance certificates in the completion-only model. Replacing each unknown size by a public class-specific upper bound gives a deployable but generally looser inequality. The homogeneous substitution recovers the main text's aggregate worst-case constant; it does not prove that each order-statistic term is separately minimal.

As an illustration, consider independently sampled truncated-Pareto sizes with $\alpha = 1.5$, $x_{\min} = 1$, hard cap 1000, sample size $N = 200$, and 5,000 samples drawn with seed 73621. The first ratio below sets L after the fact to the maximum of each finite sample; the second uses the fixed hard cap as the denominator.

k	$\mathbb{E}[\Lambda_{k-1}/((k-1)L_{\text{sample}})]$	$\mathbb{E}[\Lambda_{k-1}/((k-1)L_{\text{cap}})]$
4	0.6825	0.0444
8	0.4648	0.0263
16	0.3139	0.0161

Thus the larger ratios use $L = L_{\text{sample}} = \max_j C_j$, not the published timeout $L_{\text{cap}} = 1000$.

S7.5. Why kill-and-restart tiering does not escape the floor

Section 9 of the manuscript states that the obvious escape from the $\Omega(L)$ floor fails. Kill-and-restart tiering runs every job for a short first quantum τ and restarts the survivors, discarding the killed attempt; on n jobs of length L arriving together it leaves some job with an excess of $n\tau/k - L$ over first-come first-served, and that excess is wasted work rather than overtaking, so no budget removes it. The light-job constant does not improve either, a second-tier piece being still a non-preemptive block of length L : holding τ fixed and varying L , the worst light-job additive constant we measure grows as about $1.1L$ and is flat in τ . The mechanism is TAGS [19] with the dedicated hosts removed, and removing them is what breaks it: in TAGS the kill is a routing device and the work is redone on the next host, whereas in a judging queue the kill is terminal.

S8. Extremal Constructions and the Remaining Proofs

This section supports Section 6 of the manuscript: it gives in full the constructions and proofs whose ideas that section states, so that every constant the theory section calls exact is witnessed here. All of them are exact in integer arithmetic; every one was simulated and every intermediate quantity of every proof checked job by job, with an independent audit of work conservation and server capacity re-derived from the produced schedule. A cross-reference without an S points into the main manuscript.

The six subsections below run in the order the theory section raises them, and each is its own idea in one line. The cascade drives the workload gap of Lemma 1 of the manuscript to its ceiling geometrically: in m identical rounds, each opening with $k - 1$ long jobs of service L and then L short jobs of service 1, a static-priority policy that prefers the short ones carries $k - 1$ part-served jobs into the next round, and the carried workload obeys $\rho_{j+1} = (k - 1)(\rho_j + L)/k$, so that $\rho_m = (k - 1)L(1 - ((k - 1)/k)^m) \rightarrow (k - 1)L$; the rescaling that fixes an arbitrary service bound is part of the lemma. Both witnesses for the identity run that cascade and then inject a finale at T_m : upward, $k - 1$ long jobs, the victim and a stream of more than $2kL$ unit jobs arrive together, first-come first-served starts the victim at once with $\rho_i^{\text{FCFS}} = (k - 1)L$ while the base policy holds it until the stream runs out, leaving $\Delta_i \geq 2(k - 1)L - (k - 1)L((k - 1)/k)^m - (k - 1)$; downward, the mirrored cascade is followed by a filler job of service f , the victim and $k - 1$ jobs of service L , giving $\Delta_i = -\rho_m - (k - 1)L \rightarrow -2(k - 1)L$, and non-attainment follows from Lemma 1 of the manuscript and the *Bounding ρ* step of Theorem 1. For the guard bound, part (i) is a queue of k jobs of service L , the victim and a stream of unit jobs, where the constant budget B makes the guard fire at $L + \lceil B/k \rceil$, so the term B/k is attained; part (ii) bounds every finite instance by Theorem 3 of the manuscript strictly and approaches $3k - 2$ from below by running the cascade for m rounds and then injecting a finale timed so that $\text{over}[i]$ sits below $B = f + L + 1$ at three successive dispatch epochs, at 0, at f and at $f + L = B - 1$, the last by a single unit, which lets the base policy fill every server three times before the guard fires. The single-server family places the victim ahead of $B/g - 1$ jobs of service g and one job of service L , with g dividing B : the guard does not fire while $\text{over}[i] = B - g < B$, so the long job is taken and $\text{excess}_p[i] - B = L - g$, which approaches L and never reaches it. The price of the guarantee counts inverted pairs in two ways, which makes $\sum_i \text{In}_i$ and $\sum_i \text{Out}_i$ the two sums of service times over the inversions exactly — the pairwise-interchange argument behind Smith's rule [20] stated on an arbitrary arrival sequence — and then bounds the numerator through Corollary 2 of the manuscript and the denominator through the *mn* inversions of shortest job first. Achievability counts the short jobs exact shortest job first places before the first long one, which is $r_g = \min(n, k\lceil B/(k\sigma) \rceil)$ and at least qn , while maximality turns on one relabelling: if a wrapper permits a later-ranked job while the head has $\text{over}[h] > G - L$, giving that job the unrevealed size L leaves the transcript

the wrapper sees unchanged and pushes the head past G , after which the frontier counts
 $r = \lfloor (G - L)/\sigma \rfloor + 1$ permitted short-before-long moves.

S8.1. The Cascade

This subsection supports the claim of Section 6 of the manuscript that the constant
 $2(k-1)L$ of Theorem 1 cannot be lowered, by driving the workload gap of Lemma 1 to its
 ceiling.

Lemma 2. Fix $k \geq 2$ and $m \geq 1$ and set $L = k^m$. There is an input and a work-conserving
 policy P such that, at time $T_m = mL$, first-come first-served has completed every job and idles all k
 servers, while P holds exactly $k-1$ jobs in service, each with remaining service

$$f = \frac{\rho_m}{k-1} = L \left(1 - \left(\frac{k-1}{k} \right)^m \right), \quad \rho_m = (k-1)L \left(1 - \left(\frac{k-1}{k} \right)^m \right),$$

so that $U_P(T_m) - U_{\text{FCFS}}(T_m) = \rho_m \rightarrow (k-1)L$ as $m \rightarrow \infty$. Every ρ_j and every intermediate
 time is an integer. For any prescribed service bound $L_0 > 0$, multiplying every arrival time and
 service time in the construction by L_0/k^m gives the same ratios with bound L_0 ; integrality is used
 only before this final rescaling.

Proof. The input consists of m identical rounds; round j occupies $[T_j, T_{j+1}) = [jL, (j+1)L)$. At time T_j inject, in rank order, $k-1$ long jobs of service L and then L short jobs of
 service 1, so that the long jobs carry the lower ranks. Let P be the static-priority policy
 that dispatches a waiting short job whenever one exists and a long job otherwise; it is
 work-conserving.

The empty-system base round must be treated separately. At $T_0 = 0$ both systems
 are empty. First-come first-served dispatches the $k-1$ long jobs and one short job; its
 remaining server completes all L short jobs by $T_1 = L$, when the long jobs also complete.
 Policy P first processes the L short jobs across all k servers, finishing them at L/k , and
 then dispatches the $k-1$ long jobs. At T_1 each has L/k remaining, so $\rho_1 = (k-1)L/k =$
 $(k-1)(\rho_0 + L)/k$ with $\rho_0 = 0$.

For the induction step, fix $1 \leq j < m$ and suppose that at T_j FCFS is empty, whereas
 P has exactly $k-1$ carried-over jobs with equal remaining service and total ρ_j . FCFS again
 dispatches the new $k-1$ long jobs and one short job; one server processes the L short jobs
 and all work finishes at T_{j+1} . Under P , the one free server takes a short job. The carried-
 over jobs complete at $T_j + \rho_j/(k-1)$ and their servers then take short jobs. All k servers are
 busy until the short work runs out at $T_j + \tau_j$, where $k\tau_j = \rho_j + L$. Indeed, $\rho_j/(k-1) \leq \tau_j$ is
 equivalent to $\rho_j \leq (k-1)L$, which follows from the induction hypothesis. At that instant
 P dispatches the round's $k-1$ long jobs and leaves one server idle.

On $(T_j + \tau_j, T_{j+1}]$, FCFS has k busy servers and P has $k-1$. At T_{j+1} each of those
 long jobs has τ_j left, reproducing the hypothesis with $\rho_{j+1} = (k-1)\tau_j = (k-1)(\rho_j + L)/k$.
 Together with the base round this solves to $\rho_j = (k-1)L(1 - ((k-1)/k)^j)$. With $L = k^m$,
 every raw ρ_j and τ_j for $j \leq m$ is an integer; the rescaling in the statement is then applied
 to the whole input, including its arrival times. $\square$

S8.2. Tightness of the Net-Overtake Identity

This subsection proves Theorem 2 of the manuscript, that $\sup |\Delta_i| = 2(k-1)L$ in
 both directions.

Proof of Theorem 2 of the manuscript. *Upward.* Run the cascade of Lemma 2 for m
 rounds and, at T_m , inject in rank order $k-1$ jobs of service L , then the victim i of ser-

vice L , then $n_{\text{sh}} > 2kL$ jobs of service 1. Let P continue as the static priority “the $k - 1$ new long jobs first, then the short jobs, then i ”.

At T_m all k servers of the reference policy are free and its waiting set in rank order is the $k - 1$ new long jobs, then i , then the short ones. It dispatches the long jobs and then i , all in the phase at T_m . Hence $W_{\text{FCFS}}[i] = 0$; the $k - 1$ new long jobs are dispatched strictly before i in its sequence and are in service at $s_i^{\text{FCFS}} = T_m$ with their full service, so $\rho_i^{\text{FCFS}} = (k - 1)L$ and $R_i^{\text{FCFS}} = (k - 1)L$.

Under P , one server is free at T_m and takes a new long job; when the $k - 1$ carried-over jobs finish, all at $T_m + f$ with $f < L$, their servers take the remaining new long jobs and then short jobs. The victim waits while any short job waits. Short jobs are executed at rate at most k , so they are not exhausted before $T_m + n_{\text{sh}}/k$, and $n_{\text{sh}} > 2kL > k(f + L)$ makes that later than $T_m + f + L$, by which time every new long job has completed. So i is dispatched after all $k - 1$ new long jobs have finished, and at s_i^P the other $k - 1$ servers hold short jobs, giving $0 \leq \rho_i^P \leq k - 1$. Also $R_i^P = \rho_m + (k - 1)L$, since the carried-over jobs and the new long jobs all have rank below i and the new long jobs are untouched at a_i . Therefore $R_i^P - R_i^{\text{FCFS}} = \rho_m$ and

$$\begin{aligned}\Delta_i &= (R_i^P - R_i^{\text{FCFS}}) - \rho_i^P + \rho_i^{\text{FCFS}} \geq \rho_m + (k - 1)L - (k - 1) \\ &= 2(k - 1)L - (k - 1)L\left(\frac{k-1}{k}\right)^m - (k - 1),\end{aligned}$$

and both error terms are $o(L)$ as $m \rightarrow \infty$ with $L = k^m$.

Downward. Run the mirrored cascade: inject the L short jobs first in rank order and the $k - 1$ long jobs after, and let P be “long jobs before short ones”. The induction of Lemma 2 runs with the roles of the two policies exchanged, so at T_m the policy P is empty and idle while the reference holds $k - 1$ jobs with f of service left each. At T_m inject, in rank order, one *filler* job of service f , the victim i of service $L - 1$, and $k - 1$ jobs of service L ; let P be “the $k - 1$ new long jobs first, then i , then the filler”.

At T_m the reference policy has $k - 1$ servers busy with carried-over jobs, each with exactly f left, and one free server, which takes the filler — the lowest-ranked waiting job — which also has service f . So all k of its servers complete simultaneously at $T_m + f$, and it then dispatches i , with nothing else dispatched in that phase before it. Hence $W_{\text{FCFS}}[i] = f$ and $\rho_i^{\text{FCFS}} = 0$.

Under P the system is empty at T_m with k free servers, and it dispatches the $k - 1$ new long jobs and then i , all at T_m . So $W_P[i] = 0$, $\text{In}_i = (k - 1)L$, $\rho_i^P = (k - 1)L$, and the filler, of rank below i , is dispatched after i , so $\text{Out}_i = f$. Therefore

$$\begin{aligned}\Delta_i &= k(0 - f) - ((k - 1)L - f) = -(k - 1)f - (k - 1)L \\ &= -\rho_m - (k - 1)L \rightarrow -2(k - 1)L.\end{aligned}$$

These calculations use the raw normalisation $L = k^m$. Given a fixed target bound L_0 and tolerance ε , multiply the entire upward or downward instance by L_0/k^m . The service bound is then exactly L_0 , all waiting times and every term in Δ_i scale by the same factor, and the upward error becomes $(k - 1)L_0((k - 1)/k)^m + (k - 1)L_0/k^m$, while the downward error tends to zero. Choosing m large proves the statement with L_0 fixed.

Non-attainment. For $k \geq 2$, Lemma 1 of the manuscript gives $|R_i^P - R_i^{\text{FCFS}}| < (k - 1)L$ strictly, and the *Bounding ρ* step of the proof of Theorem 1 of the manuscript gives $0 \leq \rho_i^P, \rho_i^{\text{FCFS}} \leq (k - 1)L$; combining them in each direction gives $|\Delta_i| < 2(k - 1)L$. $\square$

For the record, the exact values of Δ_i/L produced by these two families, against the ceiling $2(k - 1)$:

k	L	m	upward Δ_i/L	downward Δ_i/L	$2(k-1)$
2	64	6	$127/64 = 1.984$	$-127/64 = -1.984$	2
3	81	4	$290/81 = 3.580$	$-292/81 = -3.605$	4
4	64	3	$303/64 = 4.734$	$-303/64 = -4.734$	6

In the executed-work convention of Remark 1 of this file the upward column is unchanged to within $o(L)$, while the downward column becomes $-63/64$, $-130/81$ and $-111/64$, values that tend to $-(k-1)L$ as m grows rather than to $-2(k-1)L$. The same-phase share of $|\Delta_i|$ is 0.00%, 0.69% and 0.00% upward and 50.39%, 55.48% and 63.37% downward, and the victim's own excess is $+3.00L$, $+3.20L$, $+3.19L$ upward and $-0.98L$, $-0.80L$, $-0.58L$ downward.

S8.3. Tightness of the Guard Bound

Section 6 of the manuscript states that search does not see the growth of the additive constant with k . A hill-climb over instances of at most 20 jobs suggests a flat value near $1.75L$, while the witness below needs 396 jobs at $k = 2$.

This subsection proves Theorem 5 of the manuscript, that both constants of the guard bound are exact.

Proof of Theorem 5 of the manuscript(i). At $t = 0$ let there arrive, in rank order, k jobs of service L , the victim i , and at least $B + k$ jobs of service 1; let the base policy be the static priority “the k long jobs, then the short ones in rank order, then i ”, and take the constant budget B . First-come first-served dispatches the k long jobs in the phase at $t = 0$, filling every server, so $W_{\text{FCFS}}[i] = L$.

If $B = 0$ then $E(t)$ contains every waiting job at every dispatch epoch and the wrapper is first-come first-served, so the excess is $0 = \lceil B/k \rceil$. If $B > 0$ then at $t = 0$ no job has completed, $E(0)$ is empty, and the base fills the servers with the k long jobs. They complete at $t = L$, and $\text{over}[i](L) = 0$ still, every completed job having rank below i . From $t = L$ on, all k servers are free at each integer instant; while $\text{over}[i] < B$ the set E is empty — the waiting jobs of rank above i have no completed job of still higher rank ahead of them — so the base dispatches k short jobs, which complete one time unit later. Hence $\text{over}[i](L + r) = rk$, the first instant at which $\text{over}[i] \geq B$ is $L + \lceil B/k \rceil$, and there i is the minimum-rank member of E and is dispatched. So $\text{excess}_P[i] = \lceil B/k \rceil$. $\square$

Proof of Theorem 5 of the manuscript(ii). For every finite instance, Theorem 3 of the manuscript gives $c_{\text{inst}}(k) < 3k - 2$: for a constant budget, $k \text{excess}_P[i] \leq \text{In}_i - \text{Out}_i + 2(k-1)L \leq \text{In}_i + 2(k-1)L < B + kL + 2(k-1)L = B + (3k-2)L$, the last step being the strict inequality (3) of the manuscript. When i has no overtaker the excess is at most $(2 - 2/k)L$ and the inequality is strict as well.

For the lower bound at $k = 1$, the family of Section S8.4 gives $\text{excess}_P[i] - B = L - g \rightarrow L$, so $(k \text{excess}_P[i] - B)/L \rightarrow 1 = 3k - 2$. For $k \geq 2$, fix $m \geq 1$, $L = k^m$, and let f and ρ_m be as in Lemma 2. Run the cascade for m rounds. At T_m inject, all arriving at T_m and in rank order,

- $k - 1$ new long jobs of service L , of rank below the victim;
- the victim i , of service L ;
- one sync job of service f ;
- one pre job of service L ;
- k final jobs of service L ;

set the constant budget to $B = f + L + 1$, and let the base policy be the static priority “short jobs of the cascade, then sync, then the new long jobs and pre, then the finals, and the victim last”.

The reference run. At T_m it is empty with all k servers free; it dispatches the $k - 1$ new long jobs and then i in the phase at T_m , so $W_{\text{FCFS}}[i] = 0$.

The guard does not fire during the cascade. Inside round j the only waiting jobs with completed higher-ranked work are that round's long jobs, whose higher-ranked jobs are the L short jobs of the round and everything in later rounds, which have not arrived. Over $[T_j, T_j + \tau_j]$ the k servers execute $k\tau_j = \rho_j + L$ units of work, of which ρ_j belongs to the carried-over jobs, so at most L units of short-job work can have completed. Hence $\text{over}[q](t) \leq L < B$ for every waiting q at every dispatch epoch of the cascade, $E(t)$ is empty throughout, and the wrapped run reproduces Lemma 2.

The finale. At T_m one server is free and $\text{over}[q](T_m) = 0$ for every waiting q , so the base chooses and takes the sync job. The $k - 1$ carried-over jobs have exactly f left and the sync job has service f , so all k servers free simultaneously at $T_m + f$. There $\text{over}[i] = f$, because only the sync job has rank above i among the completed jobs, and $f < B$; no other waiting job has a completed higher-ranked job either, so E is empty and the base fills all k servers with the $k - 1$ new long jobs and pre. These all have service L and complete simultaneously at $T_m + f + L$, where

$$\text{over}[i] = f + L = B - 1 < B,$$

since only sync and pre have rank above i among the completed jobs. So E is still empty, the base fills all k servers with the k final jobs, and they complete at $T_m + f + 2L$, at which instant $\text{over}[i] = B - 1 + kL \geq B$. The guard fires, i is the minimum-rank member of E , and it is dispatched with every server free, so $\rho_i^P = 0$.

The value. $W_P[i] = f + 2L$, $\text{In}_i = B - 1 + kL$, $\text{Out}_i = 0$, and

$$\frac{k \text{excess}_P[i] - B}{L} = \frac{k(f + 2L) - (f + L + 1)}{L} = \frac{(k - 1)f + (2k - 1)L - 1}{L}.$$

As $m \rightarrow \infty$ with $L = k^m$ we have $f/L \rightarrow 1$ and $1/L \rightarrow 0$, so the ratio tends to $(k - 1) + (2k - 1) = 3k - 2$. Hence the supremum, although no finite instance attains it, equals $3k - 2$.

The same instance also has $\Delta_i = \rho_m + (k - 1)L \rightarrow 2(k - 1)L$, so the two sources of slack in Theorem 3 of the manuscript are saturated together. $\square$

The values produced, against the ceiling $3k - 2$:

k	L	m	jobs	B	excess	$(k \text{excess} - B)/L$
2	64	6	396	128	191	$127/32 = 3.969$
2	128	7	909	256	383	$255/64 = 3.984$
3	81	4	340	147	227	$178/27 = 6.593$
3	243	5	1233	455	697	$1636/243 = 6.733$
4	64	3	211	102	165	$279/32 = 8.719$
4	256	4	1046	432	687	$579/64 = 9.047$
5	125	3	399	187	311	$1368/125 = 10.944$
6	216	3	677	308	523	$1415/108 = 13.102$

S8.4. Approaching the Single-Server Bound

This subsection proves the second claim of Proposition 3 of the manuscript, that at one server the bound $B + L$ is the exact supremum of the excess.

Proof of the second claim of Proposition 3 of the manuscript. Fix L , a budget $B > 0$, and a granularity g dividing B with $0 < g \leq L$. Take $k = 1$ and, at $t = 0$, the victim i

of rank 0, then $B/g - 1$ jobs of service g , then one job of service L ; let the base policy dispatch the jobs of service g first, then the one of service L , and never the victim. First-come first-served starts i at once, so $W_{\text{FCFS}}[i] = 0$.

Under the wrapper, after r of the jobs of service g have completed, $\text{over}[i] = rg$, which stays below B for $r \leq B/g - 1$. At the last such instant $\text{over}[i] = B - g < B$, the guard does not fire, and the base dispatches the job of service L . When it completes, $\text{over}[i] = B - g + L \geq B$ and the guard releases i . Hence $\text{excess}_P[i] = B - g + L$, so $\text{excess}_P[i] - B = L - g$, which approaches L as $g \rightarrow 0$ and never reaches it. For example $(L, B, g) = (100, 40, 1)$ gives $\text{excess}_P[i] - B = 99$, and $(64, 32, 1)$ gives 63. $\square$

S8.5. The Price of the Guarantee

This subsection proves Theorem 6 of the manuscript, which caps how much reordering can buy at a given promise, and the exact achievable fraction at one server. At $k = 1$ the error term of Theorem 1 of the manuscript vanishes identically and the sum identity is exact, so that fraction can be computed rather than bounded.

Proposition 6 (The achievable fraction at one server). *In the family of Theorem 6 of the manuscript with $k = 1$, the largest value of $(\bar{W}_{\text{FCFS}} - \bar{W}_P) / (\bar{W}_{\text{FCFS}} - \bar{W}_{\text{SJF}})$ over all non-preemptive work-conserving P with $\text{excess}_P[j] \leq G$ for every job is exactly $\min(1, \lfloor G/\sigma \rfloor / n)$.*

Proof of Theorem 6 of the manuscript. Let $\mathcal{I}(Q)$ be the set of inverted pairs of a policy Q , that is the pairs (u, v) with $u < v$ in rank and $v \prec u$. Counting the same pairs in two ways,

$$\sum_i \text{In}_i = \sum_{(u,v) \in \mathcal{I}(Q)} C_v, \quad \sum_i \text{Out}_i = \sum_{(u,v) \in \mathcal{I}(Q)} C_u,$$

exactly, for every k . This is the standard exchange argument: at $k = 1$, where Theorem 1 of the manuscript is an equality, it says that the only way to beat first-come first-served in mean wait is to invert pairs whose later-ranked job is the shorter one, and that the saving is exactly the sum of the service differences over the inversions. That is the classical pairwise-interchange argument behind Smith's rule [20], stated here on an arbitrary arrival sequence rather than on a single batch; we use it as an observation and claim nothing new in it. Write $S(Q) = \sum_i (W_{\text{FCFS}}[i] - W_Q[i])$ for the total saving against the reference and $\mathcal{E} = 2(1 - 1/k)L(m + n)$. Summing Theorem 1 of the manuscript over the $m + n$ jobs gives

$$\left| S(Q) - \frac{1}{k} \sum_{(u,v) \in \mathcal{I}(Q)} (C_u - C_v) \right| \leq \mathcal{E} \quad \text{for every } Q. \quad (1)$$

Numerator. A pair contributes positively only when u is a long job and v a short one, contributing $L - \sigma$. For a fixed long job u , the number of short jobs dispatched before it is at most In_u / σ , and Corollary 2 of the manuscript gives $\text{In}_u \leq kG + (3k - 2)L$. Summing over the m long jobs and applying (1),

$$S(P) \leq \frac{1}{k} m (L - \sigma) \frac{kG + (3k - 2)L}{\sigma} + \mathcal{E}.$$

Denominator. Sorting by service time inverts every one of the mn long-short pairs, so its pair sum is exactly $mn(L - \sigma)$, and (1) applied to it gives $S(\text{SJF}) \geq \frac{1}{k} mn(L - \sigma) - \mathcal{E}$. At $k > 1$ the denominator carries the same error as the numerator, so treating the reference saving as exact would understate $\mathcal{E}$.

Division. Writing $\mathcal{N}$ and $\mathcal{D}$ for the two leading terms $\frac{1}{k}m(L - \sigma)(kG + (3k - 2)L)/\sigma$ and $\frac{1}{k}mn(L - \sigma)$, both bounds are positive once $\mathcal{D} > \mathcal{E}$, and then

$$\frac{S(P)}{S(\text{SJF})} \leq \frac{\mathcal{N} + \mathcal{E}}{\mathcal{D} - \mathcal{E}} = \frac{kG + (3k - 2)L}{n\sigma} + \varepsilon, \quad \varepsilon = \frac{\mathcal{E}(\mathcal{N} + \mathcal{D})}{\mathcal{D}(\mathcal{D} - \mathcal{E})}.$$

The leading terms are $\Theta(mn)$ and $\mathcal{E} = \Theta(m + n)$, so $\varepsilon = O((m + n)/(mn)) \rightarrow 0$; since $\mathcal{E}/\mathcal{D} = 2(k - 1)L(m + n)/(mn(L - \sigma))$ and $\mathcal{N}/\mathcal{D} = (kG + (3k - 2)L)/(n\sigma)$, the product $\varepsilon mn/(m + n)$ equals

$$\frac{2(k - 1)L}{L - \sigma} \left(1 + \frac{kG + (3k - 2)L}{n\sigma}\right) / \left(1 - \frac{\mathcal{E}}{\mathcal{D}}\right),$$

which exceeds the same expression without the last factor and approaches it as m, n grow. With k, L, σ, G fixed it tends to $2(k - 1)L/(L - \sigma)$, and it vanishes at $k = 1$. Dividing numerator and denominator by $m + n$ turns the two total savings into the two mean savings and leaves the ratio unchanged. $\square$

Proof. Write r_u for the number of short jobs dispatched before a long job u . Every short job ranks above every long one, so Out_u contains only long jobs, and a long–long inversion (p, q) puts L into In_p and the same L into Out_q ; summing over the long jobs those terms cancel, and since $k = 1$ makes Theorem 1 of the manuscript an equality,

$$\sum_{u \text{ long}} \text{excess}_P[u] = \sum_{u \text{ long}} (\text{In}_u - \text{Out}_u) = \sigma \sum_{u \text{ long}} r_u.$$

Let u^* be the long job dispatched last among the long ones. It has no lower-ranked long job after it, so $\text{Out}_{u^*} = 0$ and $\text{excess}_P[u^*] \geq \sigma r_{u^*}$; and r is non-decreasing along the dispatch order, so $r_{u^*} = \max_u r_u$. The guarantee therefore forces $\max_u r_u \leq \lfloor G/\sigma \rfloor$ and $\sum_u r_u \leq m \lfloor G/\sigma \rfloor$. Only long–short inversions contribute to the saving, each contributing $L - \sigma$, so $S(P) = (L - \sigma) \sum_u r_u$ against $S(\text{SJF}) = (L - \sigma)mn$ and the fraction is at most $\min(1, \lfloor G/\sigma \rfloor/n)$.

For achievability, put $r = \min(n, \lfloor G/\sigma \rfloor)$ and dispatch the r lowest-ranked short jobs, then the m long jobs in rank order, then the remaining short jobs in rank order. Every long job then has excess exactly $r\sigma \leq G$, every short job has excess at most 0, and the fraction is r/n . $\square$

So the shape of Theorem 6 of the manuscript is right at $k = 1$ and its additive $(3k - 2)L$ is slack: the guarantee is spent on the long jobs together, not one at a time. Exhaustive enumeration of every work-conserving schedule of the family, over all 1,161 parameter settings with $m \leq 3$, $n \leq 4$ and $G \leq 3L$, agrees with Proposition A1 of the manuscript in every case.

S8.6. Achievability and Maximal Permission

This subsection proves Theorem 7, Theorem 8 and Corollary 4 of the manuscript, which match the price bound on the two-size batch family and identify the pointwise-maximal permission rule at one server.

Proof of Theorem 7 of the manuscript. For a policy Q , let

$$I(Q) = \sum_{(u,v) \in \mathcal{I}(Q)} (C_u - C_v).$$

The summed identity (1) gives $|\Delta(Q) - I(Q)/k| \leq E$ and, because SJF inverts all mn long–short pairs, $D_0 - E \leq \Delta(\text{SJF}) \leq D_0 + E$.

If $B = 0$, the guard is FCFS and the lower bound is immediate because $q = 0$. Suppose $B > 0$. Until the first long job is dispatched, every completed overtaker is short and exact SJF continues to propose short jobs. They complete in waves of at most k , so the number dispatched before that first long job is

$$r_g = \min\left(n, k \left\lceil \frac{B}{k\sigma} \right\rceil\right), \quad \frac{r_g}{n} \geq q.$$

Those r_g short jobs precede every long job, and hence contribute at least $mr_g(L - \sigma)$ to $I(\text{guard})$. It follows that $\Delta(\text{guard}) \geq qD_0 - E$. Therefore

$$\Delta(\text{guard}) - q\Delta(\text{SJF}) \geq -E(1 + q),$$

and division by the positive lower bound $D_0 - E$ gives $F(\text{guard}) \geq q - \varepsilon_-$.

For any feasible P , Corollary 2 of the manuscript bounds the number of short jobs inverted ahead of each long job by $(kG + (3k - 2)L)/\sigma$. Consequently its positive inversion sum is at most AkD_0 , and (1) gives $\Delta(P) \leq AD_0 + E$. Dividing by $\Delta(\text{SJF}) \geq D_0 - E$ yields

$$F(P) \leq \frac{AD_0 + E}{D_0 - E} = A + \frac{E(1 + A)}{D_0 - E} = A + \varepsilon_+.$$

SJF minimises total waiting time for a simultaneous batch on identical servers, so $F(P) \leq 1$. The guarded policy is itself feasible by Corollary 3 of the manuscript, which places it below $F^*(G)$ and completes the sandwich. $\square$

Proof of Theorem 8 of the manuscript. At one server Theorem 1 of the manuscript is the exact relation $\text{excess}_P[i] = \text{In}_i - \text{Out}_i$. If $B = 0$, the queue head always fires and the guard is FCFS. Let $B \geq 1$. Immediately before the last overtaker of a job is dispatched, completed overtaking work is an integer strictly below B , hence at most $B - 1$; the last unrevealed job has size at most L . Thus every excess is at most $B + L - 1$. Equality is attained by putting a queue-head victim before $B - 1$ unit overtakers and then one overtaker of size L , and having the opaque base propose them in that order. This proves the first claim and shows that a positive constant budget has promise G exactly when $B \leq G - L + 1$.

For pointwise maximality, consider the minimum-rank waiting job h at a reachable dispatch history and write $w = \text{over}[h]$. All lower-ranked jobs have already been dispatched, so eventually $\text{Out}_h = 0$. If a wrapper permits a later-ranked job j when $w > G - L$, relabel the still-unfinished j to have size L and replay the same opaque proposal transcript. The wrapper has seen the same arrivals, dispatches, completions, revealed sizes and proposals, so determinism makes it take the same action. The FCFS start of h is unchanged because j has later rank, whereas $\text{excess}[h] = \text{In}_h \geq w + L > G$, contradicting the promise. Conversely, if $w \leq G - L$, permitting one later-ranked waiting job can add at most L to the head's overtaking work, so dispatching the head next remains within G . On the integer grid, $w > G - L$ is equivalent to $w \geq B^* = G - L + 1$, exactly the guard's trigger. $\square$

Proof of Corollary 4 of the manuscript. With $B^* = G - L + 1$, exact SJF proposes short jobs until the guard fires. The number that can pass every long job is

$$r = \min\left(n, \left\lceil \frac{B^*}{\sigma} \right\rceil\right) = \min\left(n, \left\lfloor \frac{G - L}{\sigma} \right\rfloor + 1\right).$$

Each of the m long jobs is inverted with exactly these r short jobs, so the closed fraction is r/n . Once their completed work exceeds $G - L$, the next long job is the queue head and Theorem 8 of the manuscript forces every opaque-base wrapper with promise G to

dispatch it; the same applies successively to the remaining long jobs. No such wrapper can therefore permit more of the short-before-long moves on this transcript. $\square$

S9. Extended Related Work

Section 2 of the manuscript keeps one paragraph per theme and a compact citation list. This section holds the per-paper detail that paragraph compresses, in the same order; every work named here is cited in the manuscript as well. Nothing in it is a result of ours.

S9.1. Fairness Referenced to First-Come First-Served

Definition 2 of Schwiegelshohn and Yahyapour [14] calls a schedule λ -fair when jobs submitted after job i cannot increase i 's flow time by more than a factor λ . Their Lemma 4 bounds fairness and efficiency jointly: the preemptive PFCFS policy is $(2 + 2\bar{s})$ -fair, with competitive bounds for weighted completion time and makespan, where $\bar{s}$ is the normalised preemption penalty, and the proof bounds the delay caused to i by a later wide job by $(1 + 2\bar{s})p_i$. That is tighter for a short job than an L -denominated bound, and it is obtained by gang preemption, which our model does not have.

In the fair-start-time line, Sabin, Kochhar and Sadayappan [21] compare a job with a re-simulation excluding later arrivals, and Sabin and Sadayappan [22] and Rajbhandary, Bunde and Leung [23] use the resulting excess as a job-level metric. The fair start time is defined in two variants; the reference used in the manuscript, first-come first-served with no backfilling, is the relaxed one.

S9.2. Overtaking Restrictions in Queueing Theory

Nudge [8], Nudge-K [9] and Nudge*(M) [24] are analysed in steady state under Poisson arrivals with light-tailed or phase-type sizes, and yield, for suitable parameters, stochastic dominance over first-come first-served. Because a job is passed at most once, Nudge implies on every sample path a per-job additive bound of one small-job length; that implication is our own reading and not a claim of [8]. Theorem 2 of the WCFS finite-skip work [10] brackets the mean response time within additive constants of the M/G/1 first-come first-served expression at every load. Yu and Scully [25] define the *crossing work* of later arrivals that precede a tagged job under γ -Boost and derive stationary response-time bounds, and Yu, Harlev, Adakroy and Scully [26] extend the line to k servers.

S9.3. The Additive Shape, Stated as Its Sources State It

Worst-case fair weighted fair queueing bounds a session's service in both directions against the ideal fluid reference, falling behind it by at most one maximum packet and running ahead of it by less than one of the session's own [27]. The error term in the packet scale rate guarantee for expedited forwarding is the transmission time of one maximum-size lower-priority packet, and exists because that packet cannot be preempted [28]; additive per-individual bounds against a reference discipline are older still [29,30]. EEVDF, a preemptive proportional-share scheduler, bounds a client's lag against the fluid ideal by $-r_{\max} < \text{lag} < \max(r_{\max}, q)$, with $r_{\max}$ the largest request and q the quantum, and by $\pm q$ when no request exceeds a quantum [31]. In Graham's list-scheduling bound the additive form is $(1 - 1/k)$ times the length of the last task to finish, and it appears inside his proof [32]. The pathwise work-difference bound nearest to Lemma 1 of the manuscript is [33, Lemma 2], where the work of jobs of remaining size at most x under k -server preemptive shortest-remaining-processing-time exceeds its single-server counterpart by at most kx ; Lemma 1 differs in comparing two arbitrary non-preemptive work-conserving k -server policies, in bounding total unfinished work, and in being two-sided. In the conservation-law family a weighted sum of mean waiting times is invariant across a class of disciplines in steady state [34], and the multiserver accumulating priority queue orders service by a

credit linear in waiting time, with exact class waiting-time distributions under exponential service [35,36].

S9.4. Deployed Systems and Relational Benchmarks

The judging-queue scheduler of Zhang, Wu and Lu [37] cuts the queue into fixed-size windows and dispatches in ascending predicted runtime within each; it reports mean response time against first-come first-served, no per-job bound and no adversarial-prediction test in its abstract or its results, and does not describe its arrival process as recorded. The tail-aware inference scheduler [38] argues about starvation in the crossing-work terms of [25]. SFS orchestrates the Linux first-in first-out and fair schedulers from user space to approximate shortest-remaining-time-first, trading longer durations for long functions against the short ones [39], and ALPS derives its per-function rank and time-slice cap from an offline shortest-remaining-processing-time replay of recent traces and reuses the fair scheduler to keep long functions from starving [40]. In the RelBench study against a human data scientist [41], the relational model matched or beat the expert on 11 of 15 tasks, while the authors record that regression tasks are where it struggles; the successor benchmark [42] keeps the same baseline construction.

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